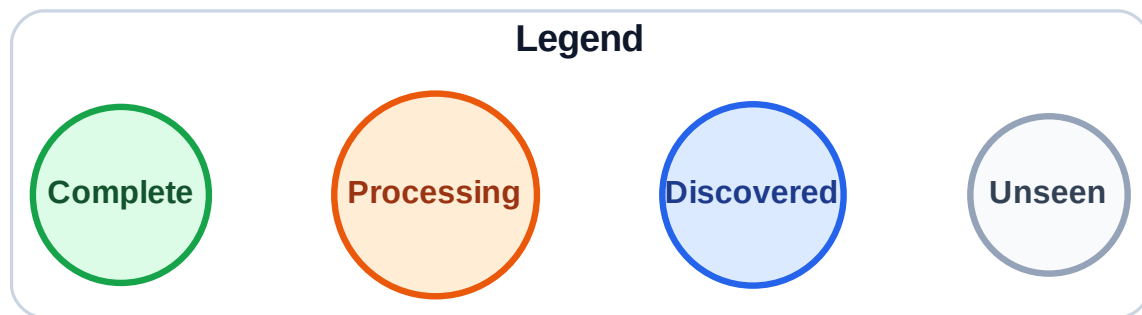
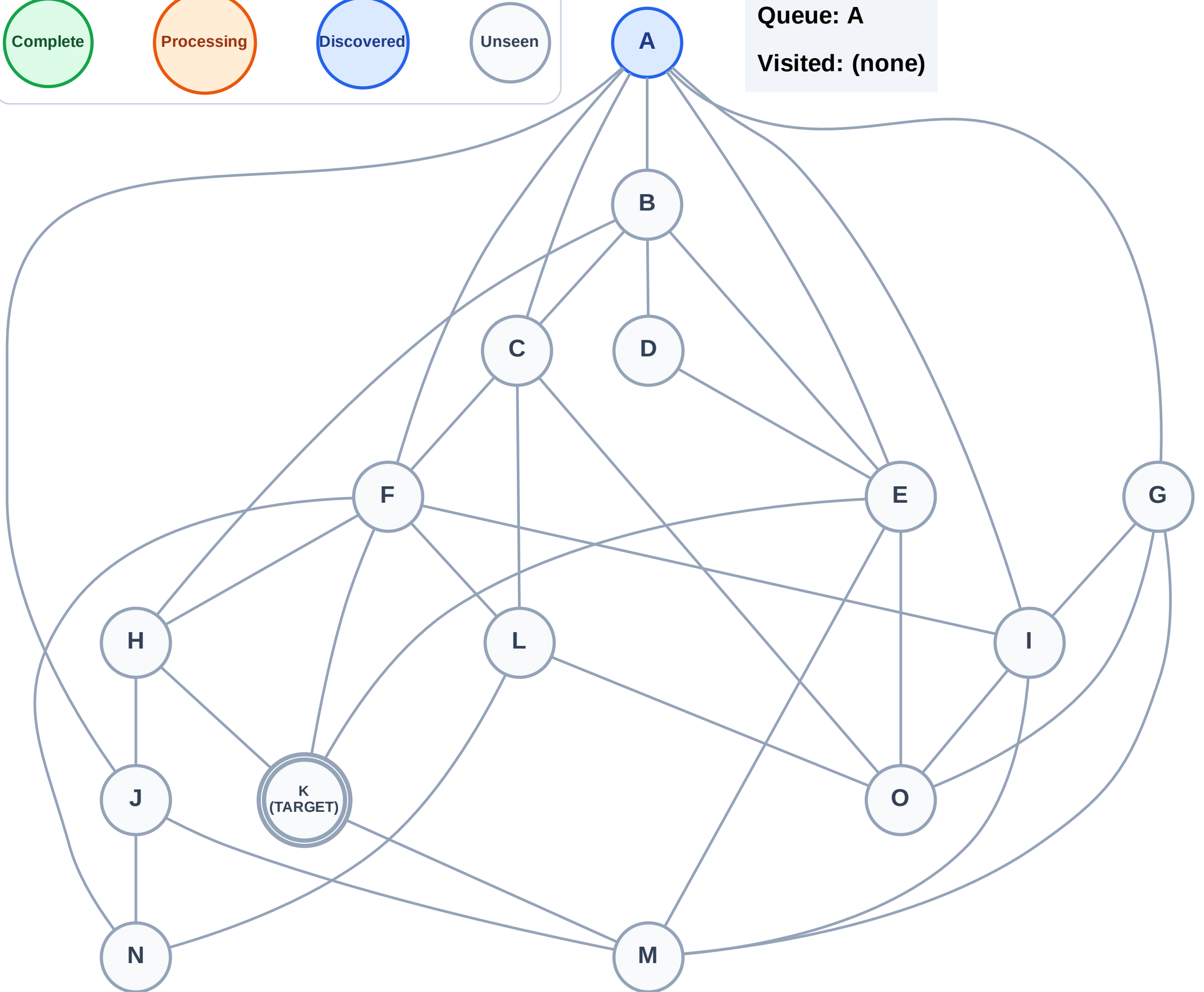


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

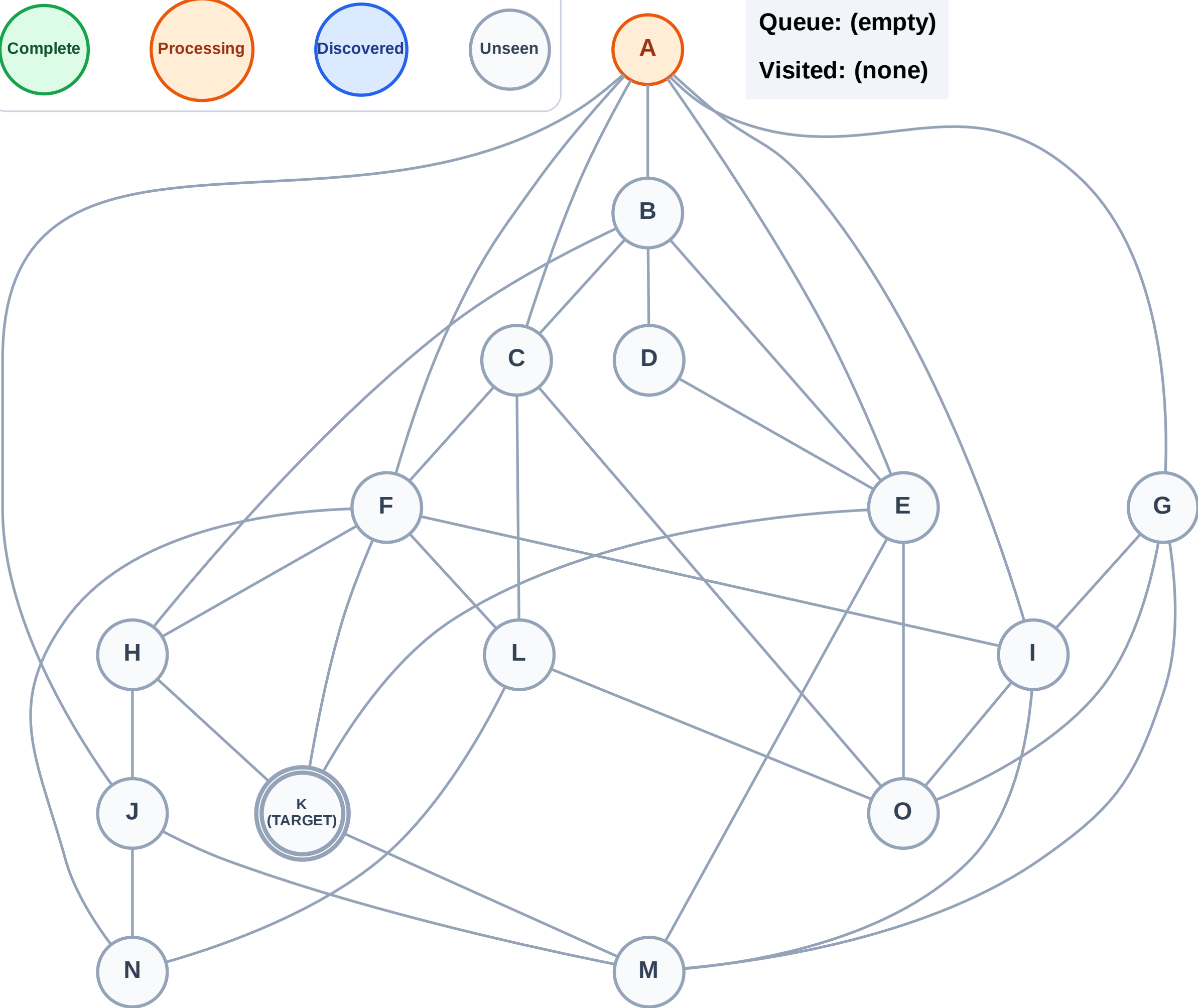
Complete

Processing

Discovered

Unseen

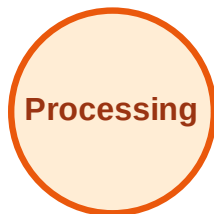
Queue: (empty)
Visited: (none)



BFS Traversal

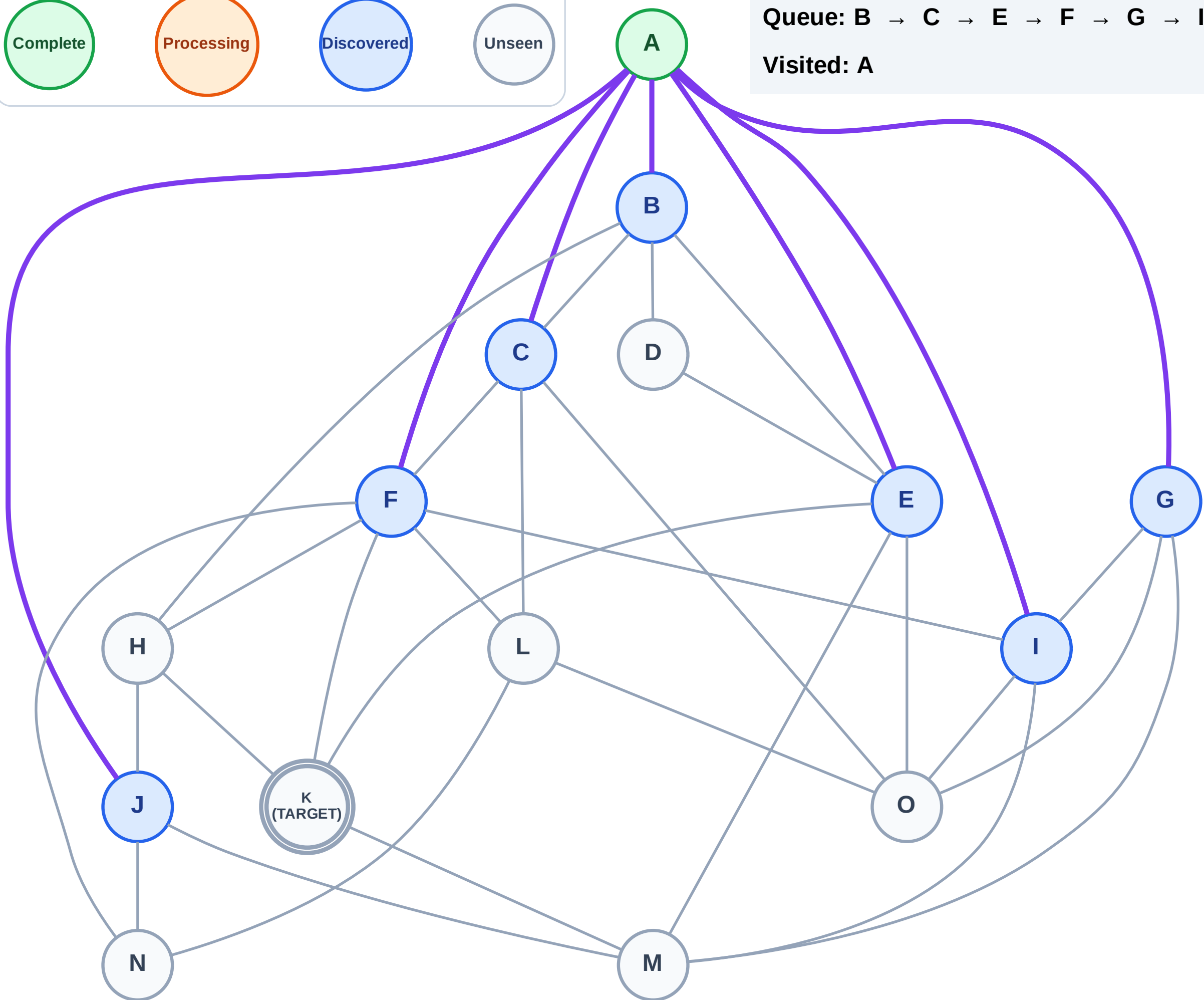
Step 3: Discover B, C, E, F, G, I, J from A

Legend



Queue: B → C → E → F → G → I → J

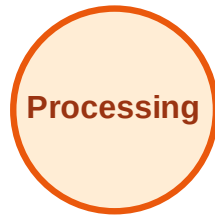
Visited: A



BFS Traversal

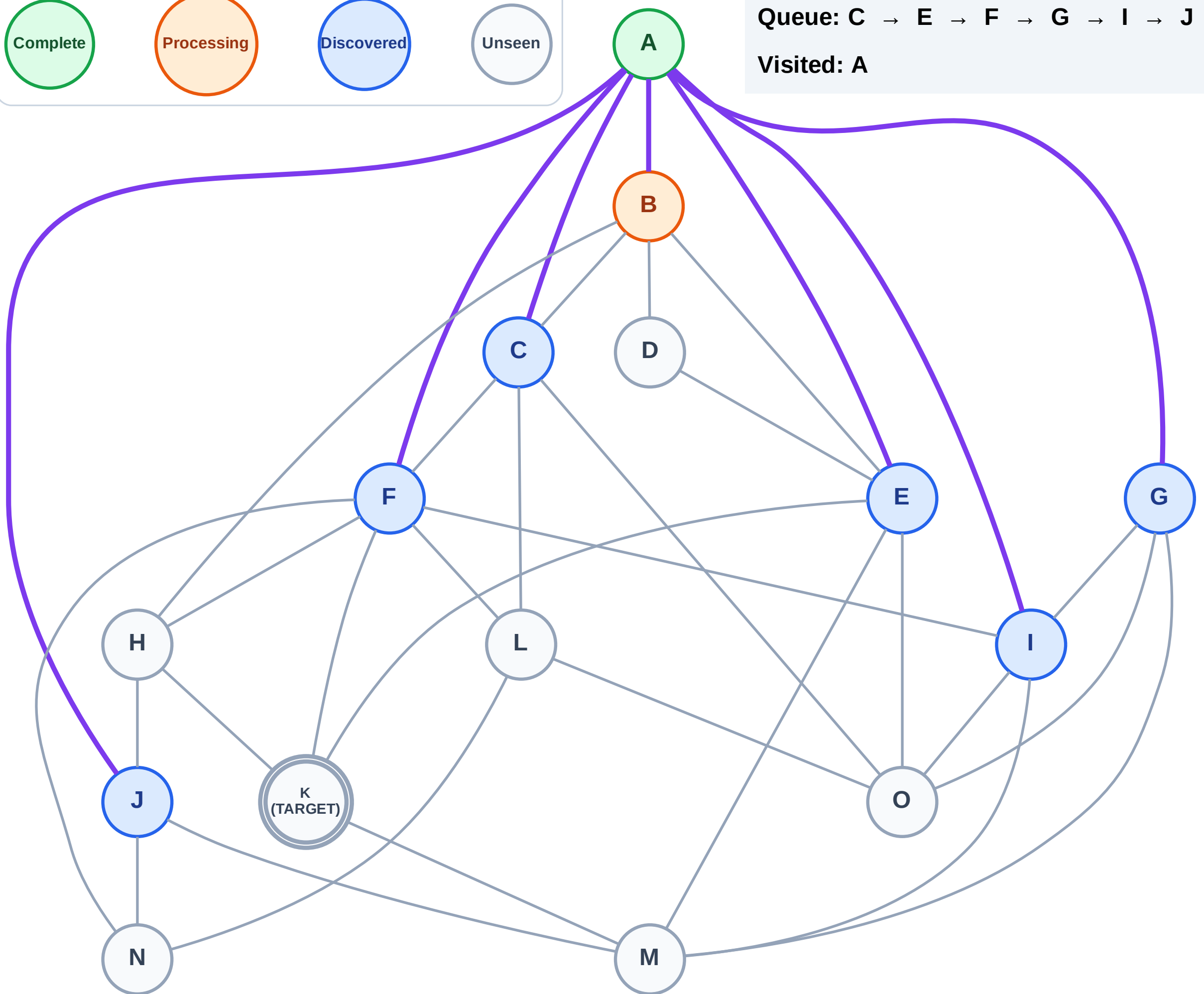
Step 4: Process node B

Legend



Queue: C → E → F → G → I → J

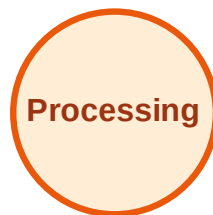
Visited: A



BFS Traversal

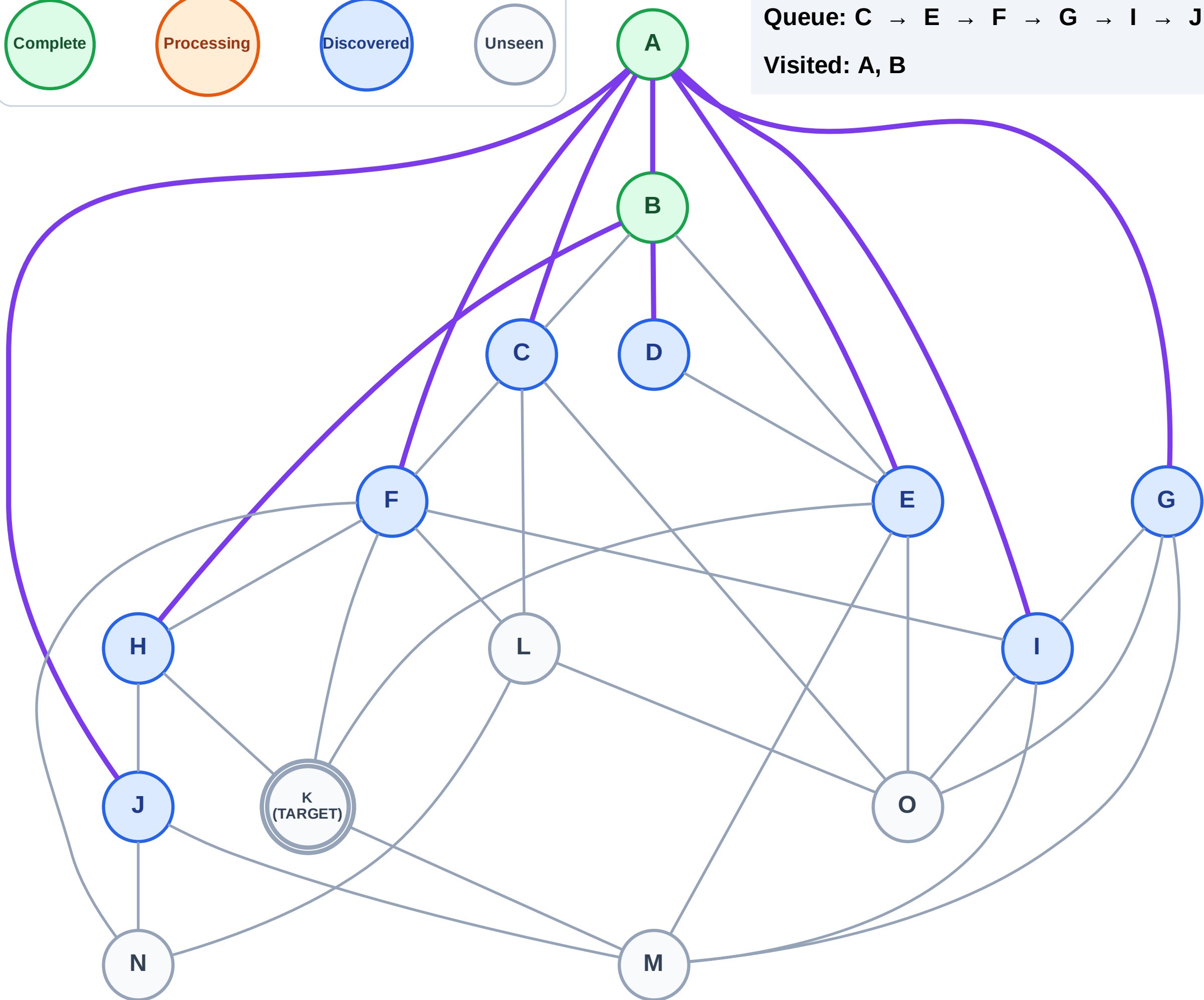
Step 5: Discover D, H from B

Legend

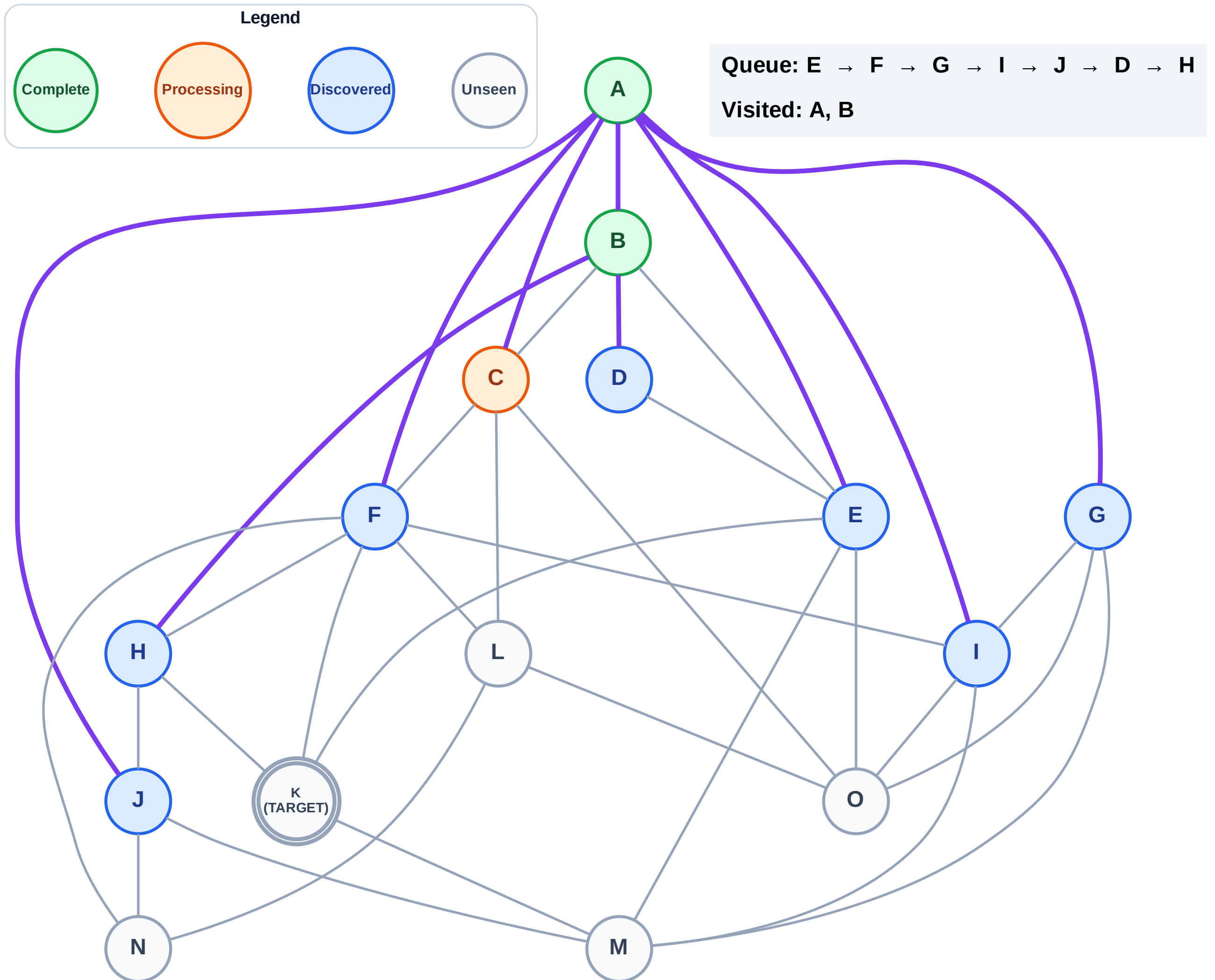


Queue: C → E → F → G → I → J → D → H

Visited: A, B



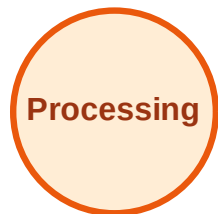
Step 6: Process node C



BFS Traversal

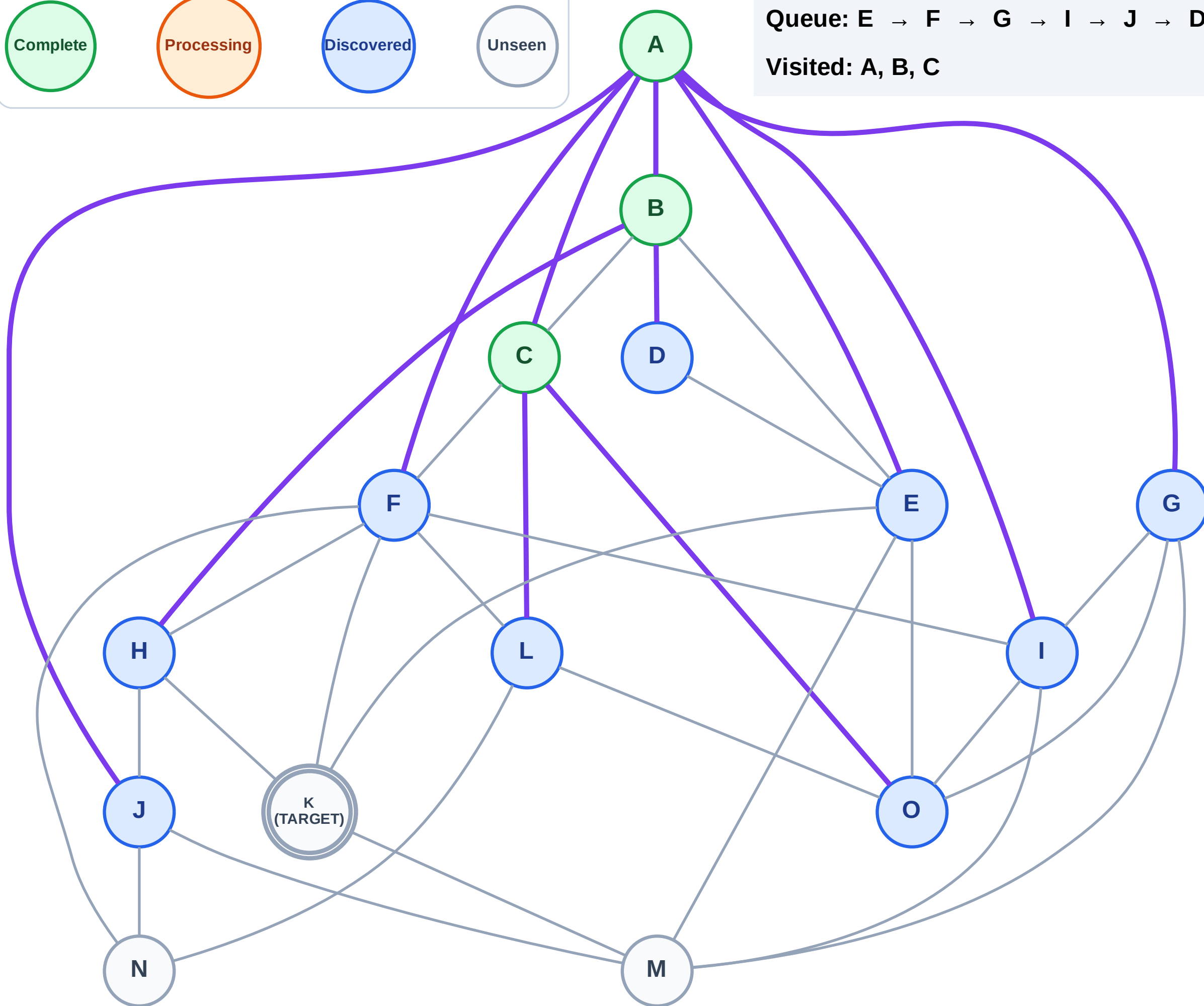
Step 7: Discover L, O from C

Legend



Queue: E → F → G → I → J → D → H → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

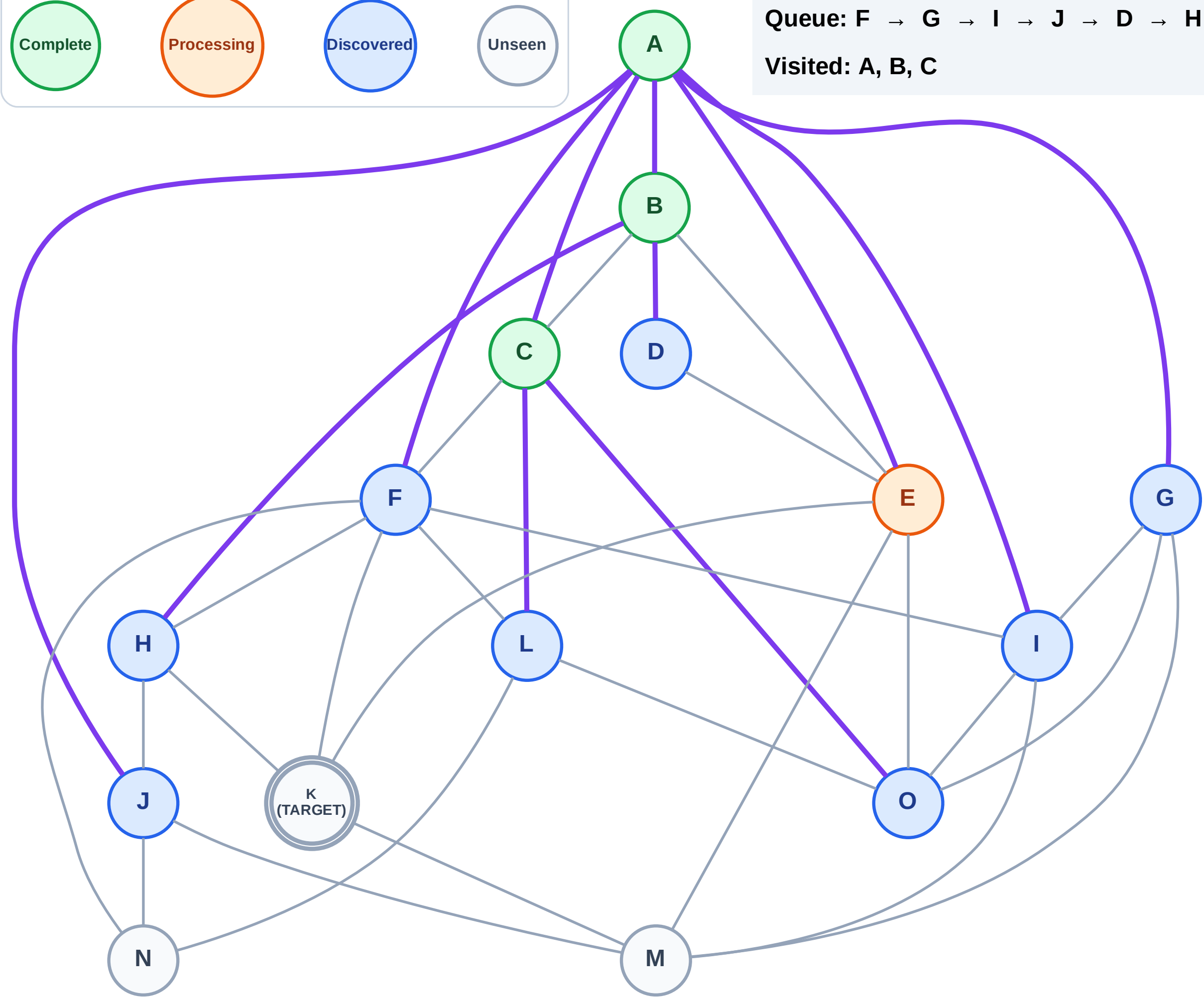
Processing

Discovered

Unseen

Queue: F → G → I → J → D → H → L → O

Visited: A, B, C

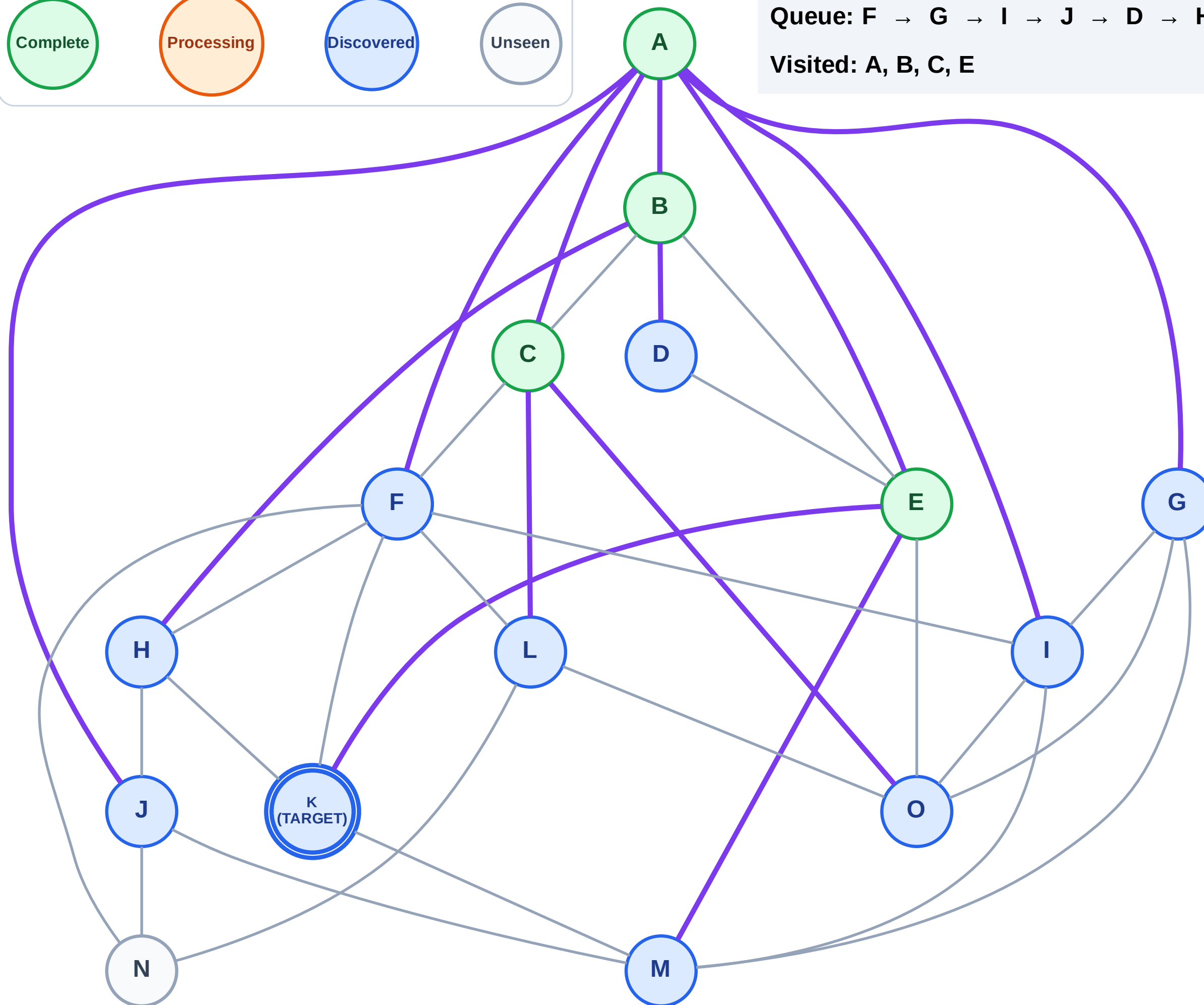
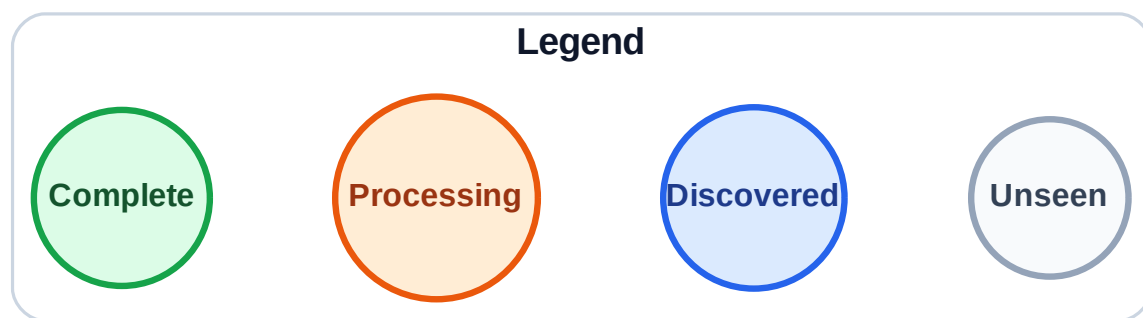


BFS Traversal

Step 9: Discover K, M from E

Queue: F → G → I → J → D → H → L → O → K → M

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

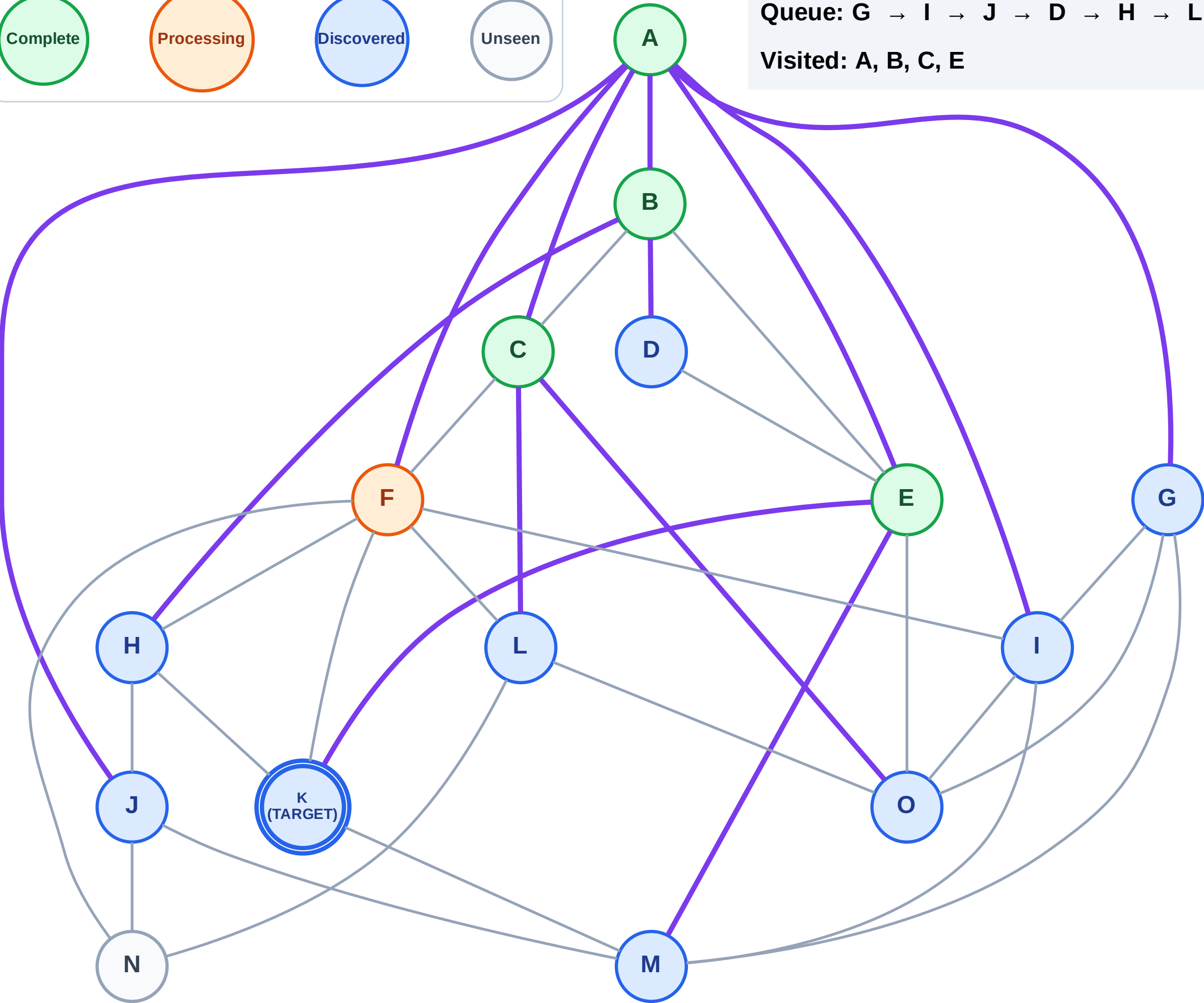
Processing

Discovered

Unseen

Queue: G → I → J → D → H → L → O → K → M

Visited: A, B, C, E



BFS Traversal

Step 11: Discover N from F

Legend

Complete

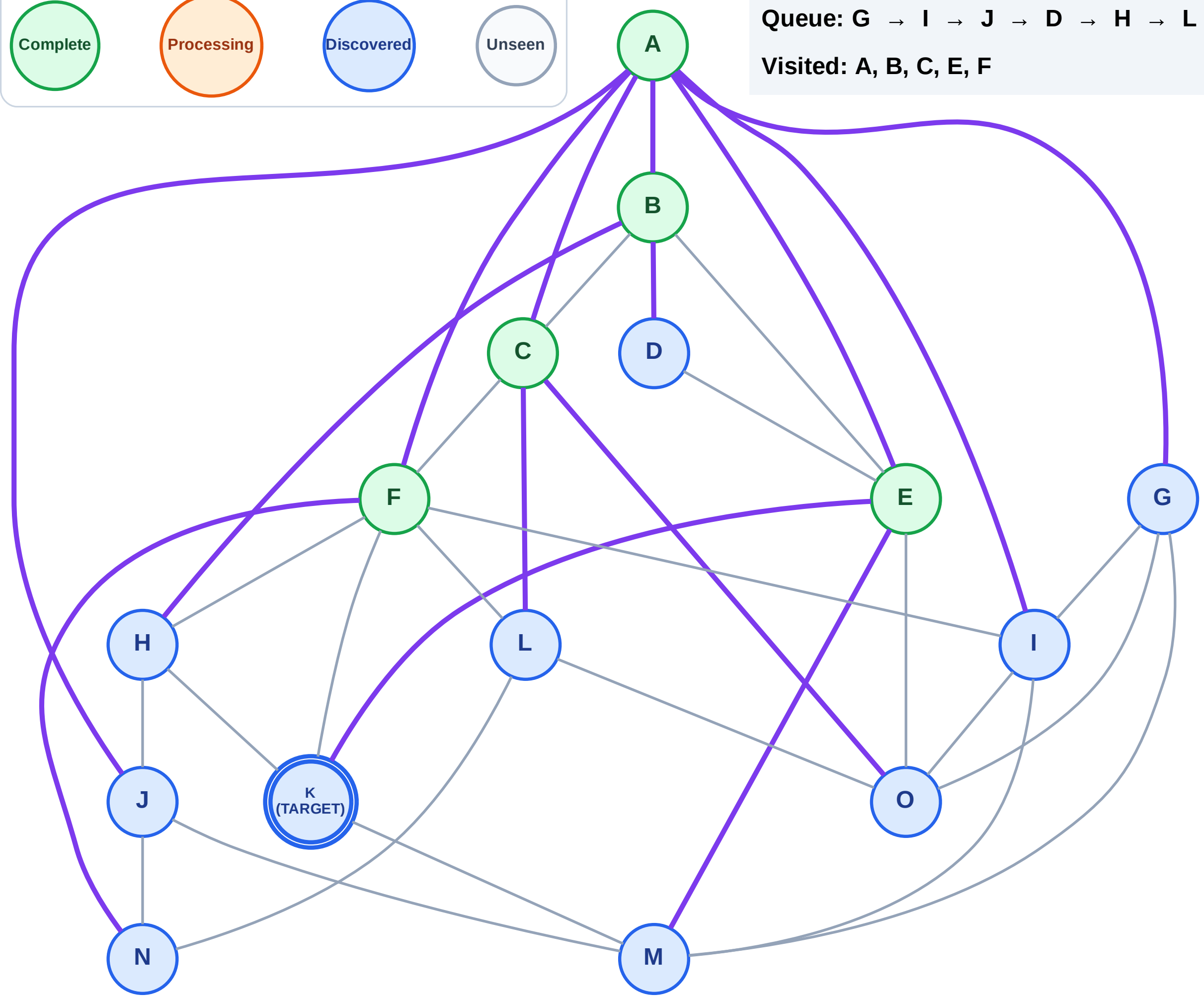
Processing

Discovered

Unseen

Queue: G → I → J → D → H → L → O → K → M → N

Visited: A, B, C, E, F



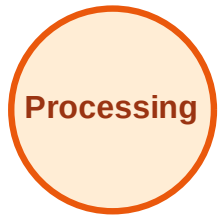
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



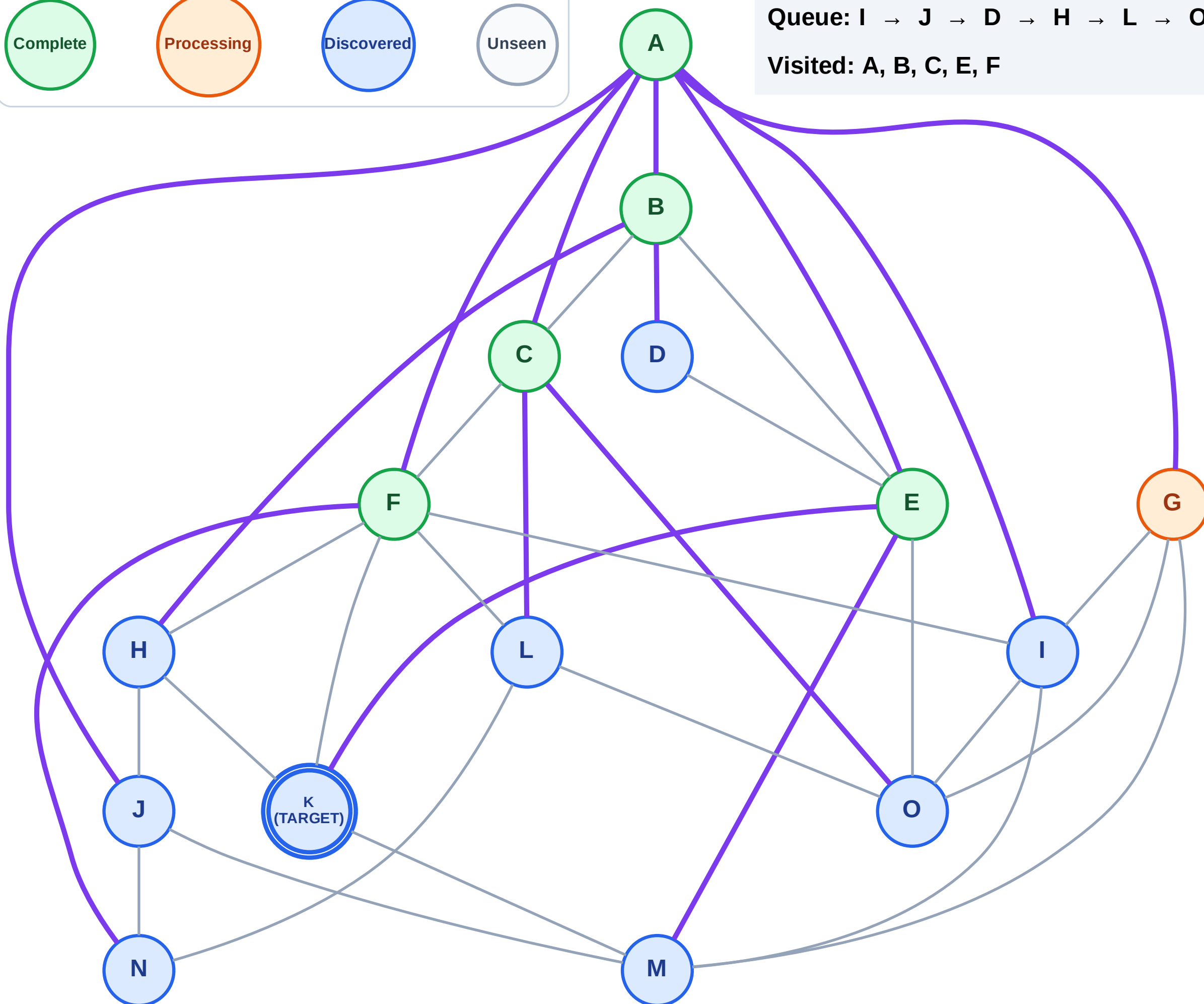
Discovered



Unseen

Queue: I → J → D → H → L → O → K → M → N

Visited: A, B, C, E, F



BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

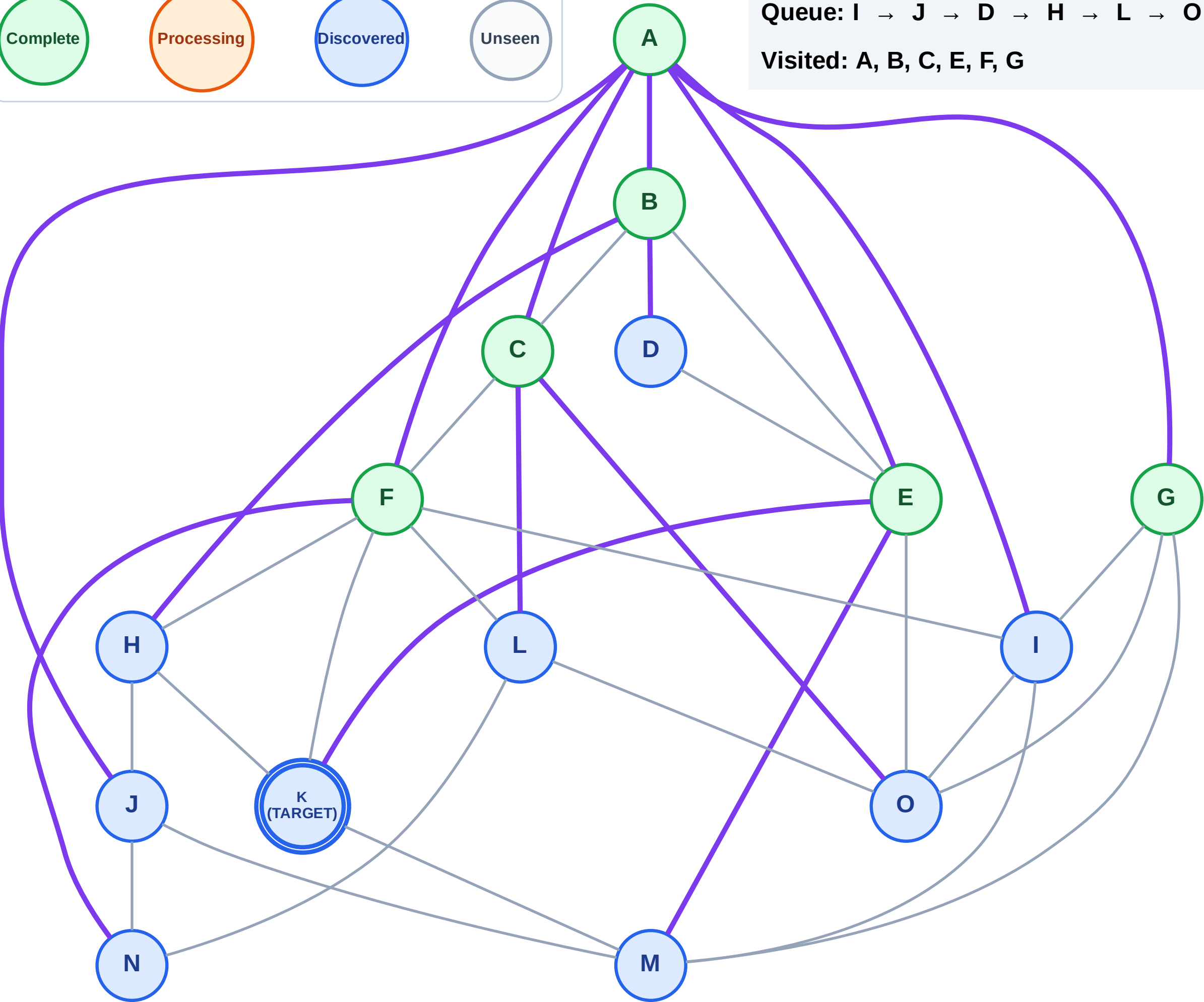
Processing

Discovered

Unseen

Queue: I → J → D → H → L → O → K → M → N

Visited: A, B, C, E, F, G



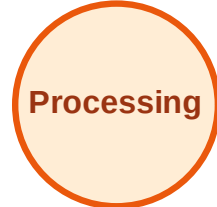
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



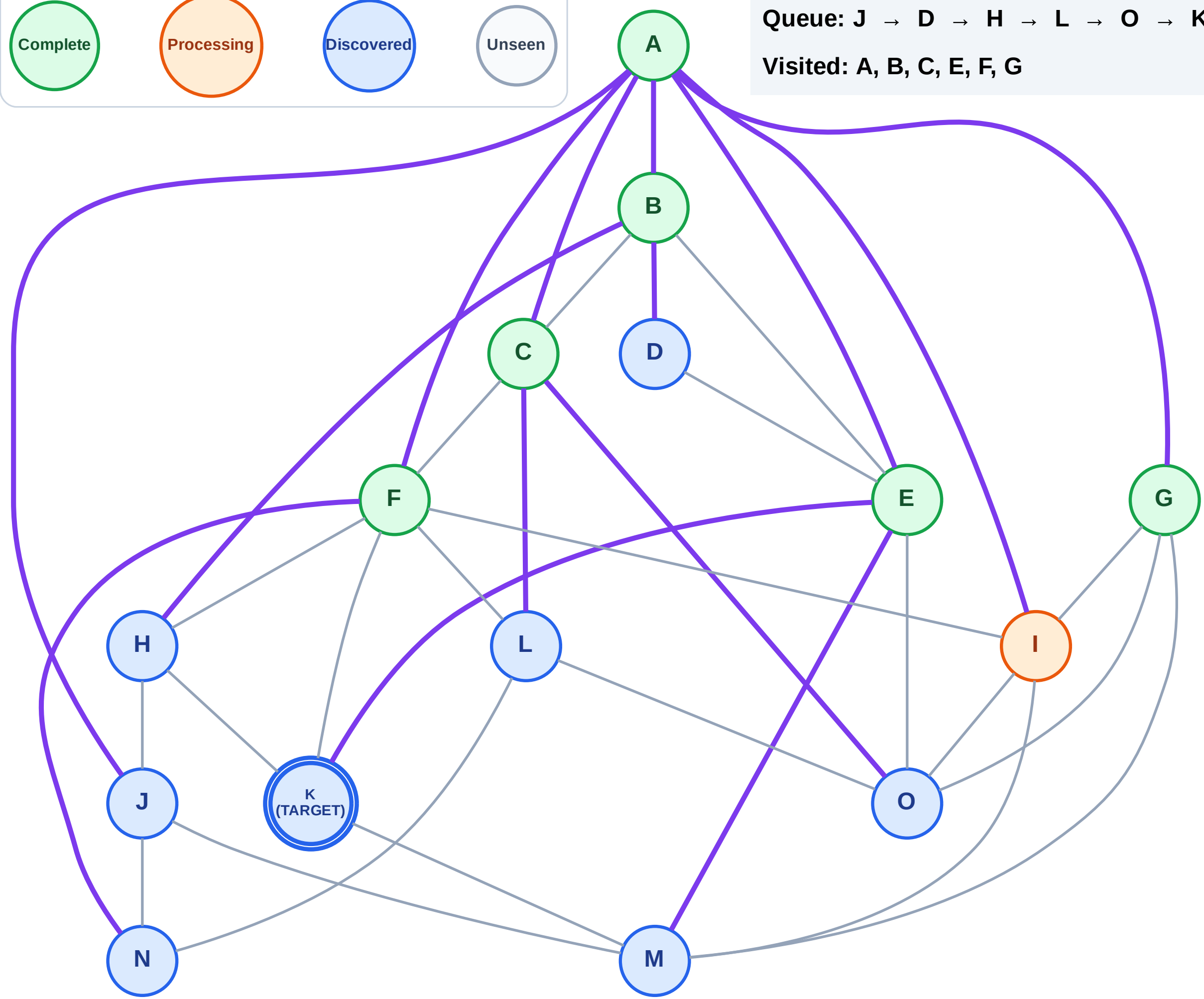
Discovered



Unseen

Queue: J → D → H → L → O → K → M → N

Visited: A, B, C, E, F, G



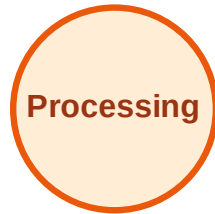
BFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



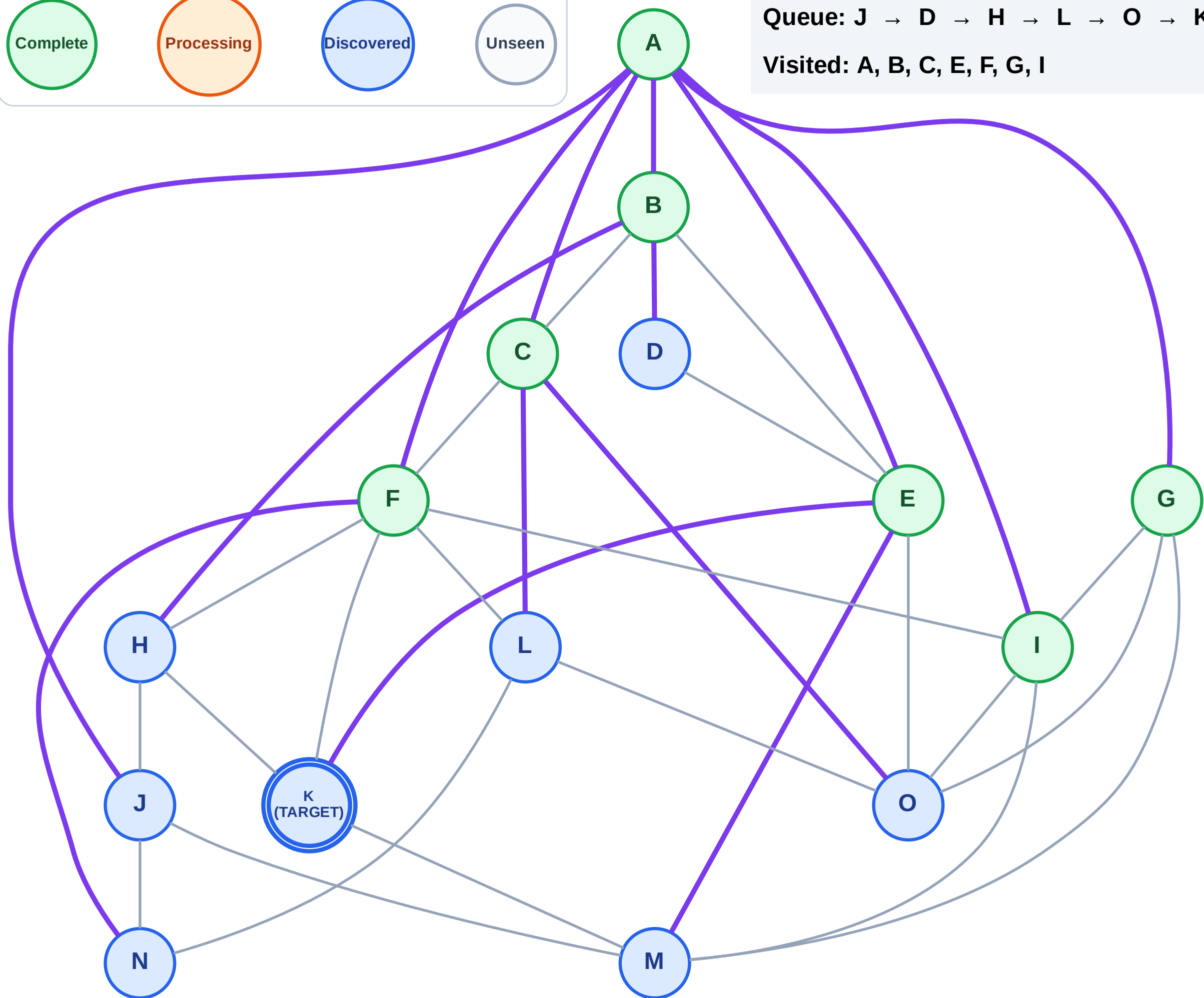
Discovered



Unseen

Queue: J → D → H → L → O → K → M → N

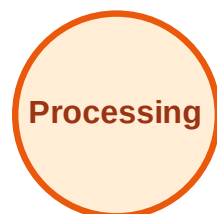
Visited: A, B, C, E, F, G, I



BFS Traversal

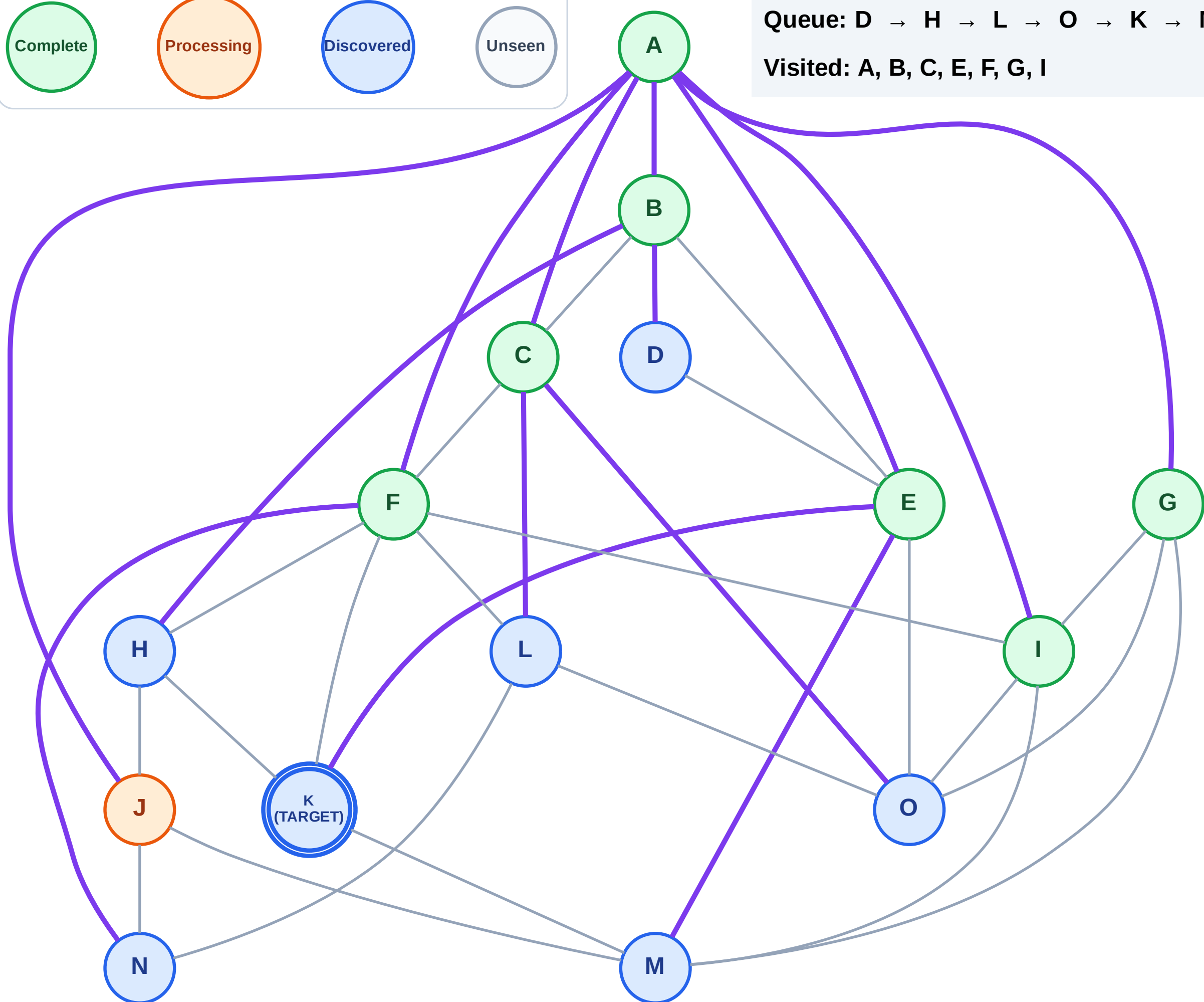
Step 16: Process node J

Legend



Queue: D → H → L → O → K → M → N

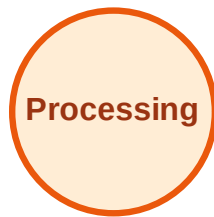
Visited: A, B, C, E, F, G, I



BFS Traversal

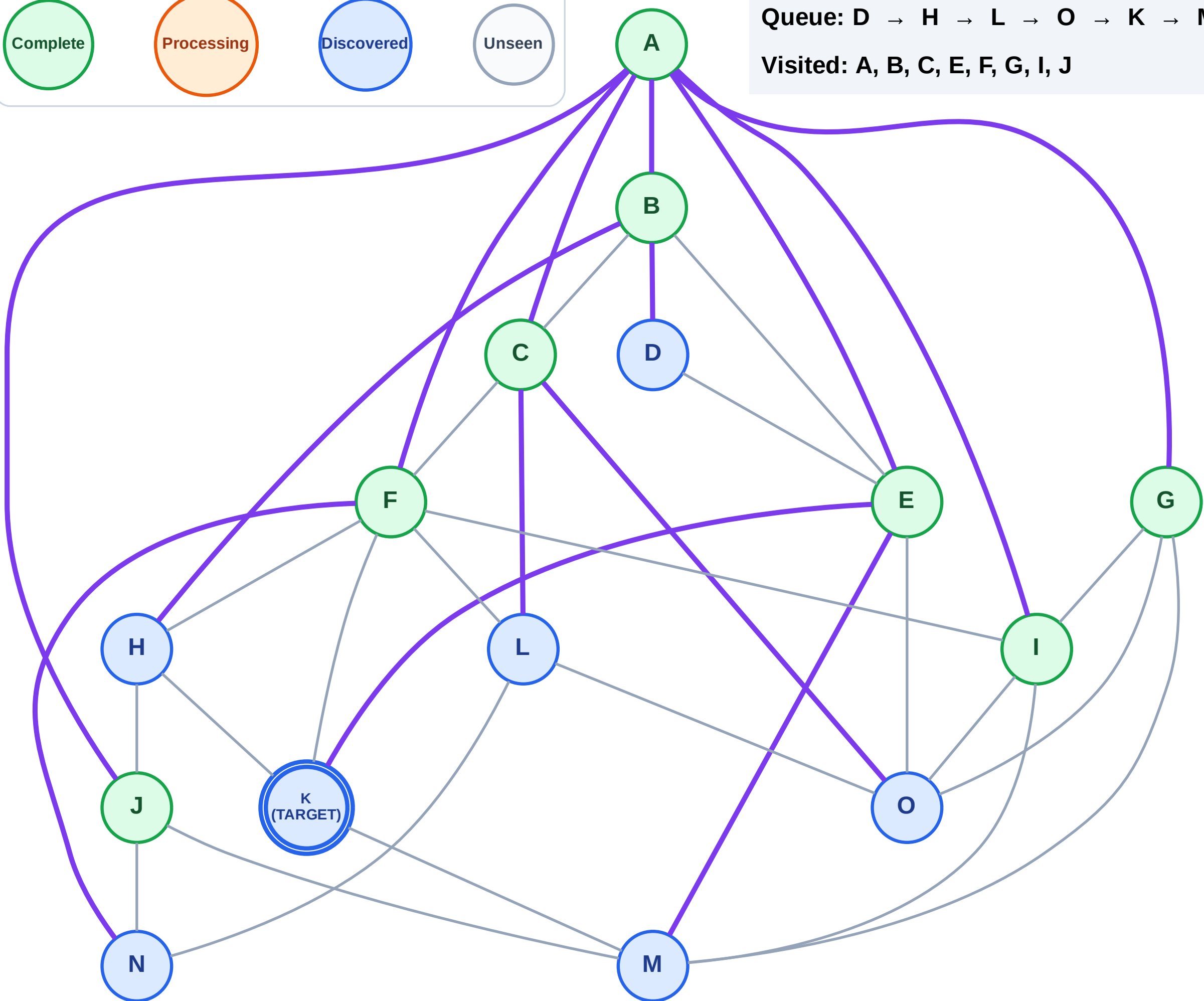
Step 17: No new neighbors from J

Legend



Queue: D → H → L → O → K → M → N

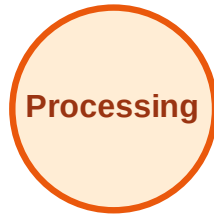
Visited: A, B, C, E, F, G, I, J



BFS Traversal

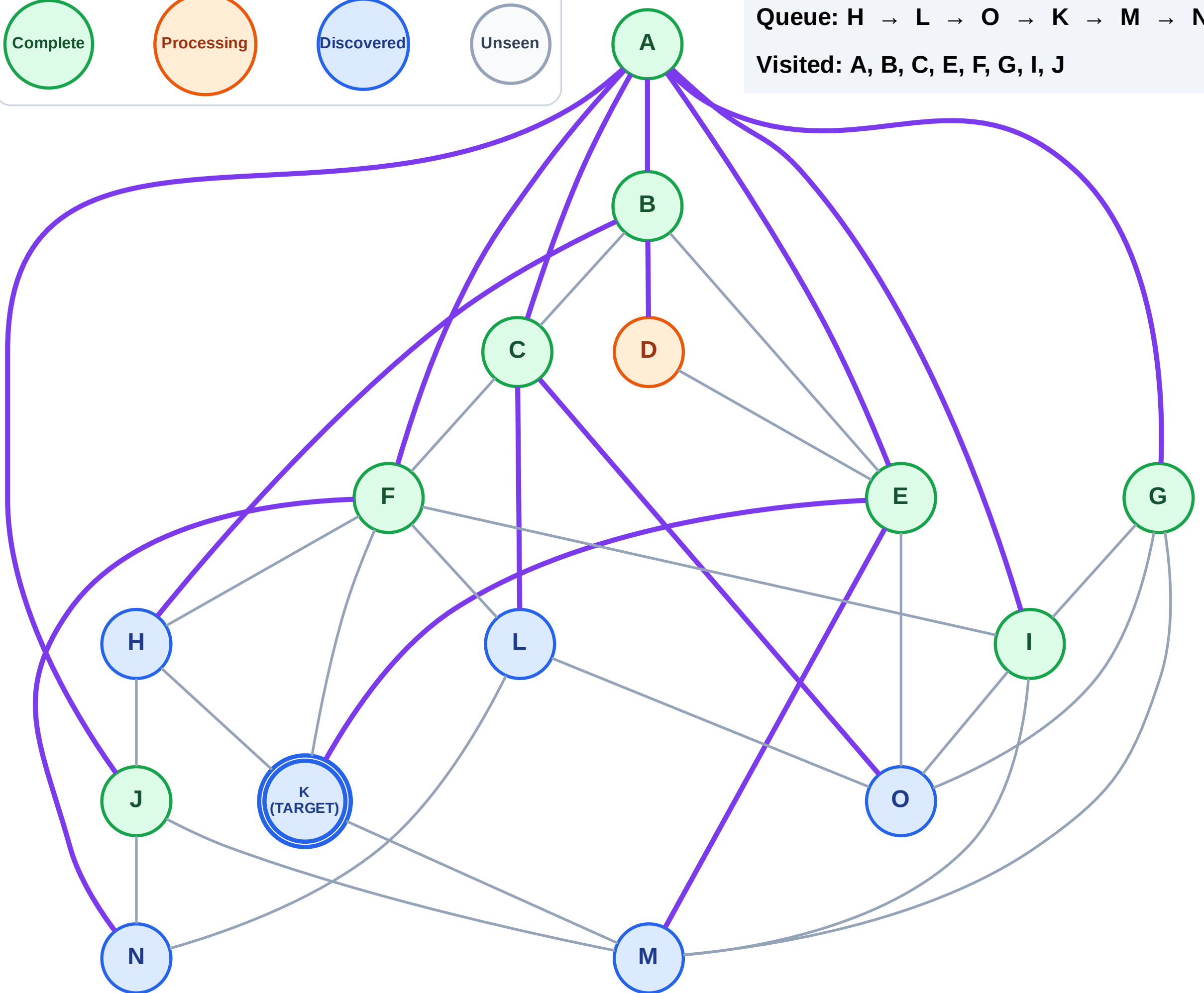
Step 18: Process node D

Legend



Queue: H → L → O → K → M → N

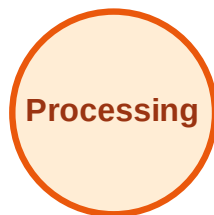
Visited: A, B, C, E, F, G, I, J



BFS Traversal

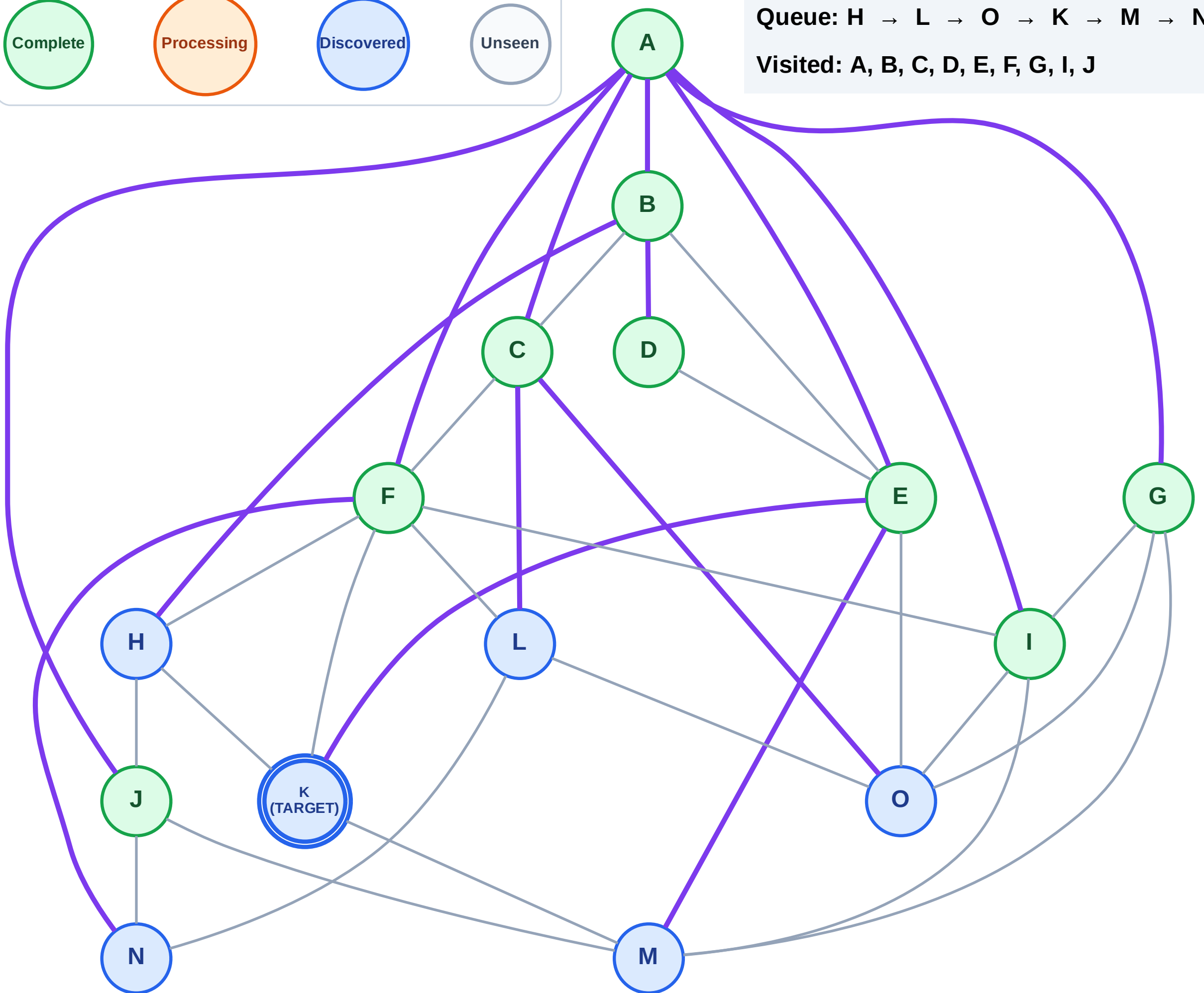
Step 19: No new neighbors from D

Legend



Queue: H → L → O → K → M → N

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

Complete

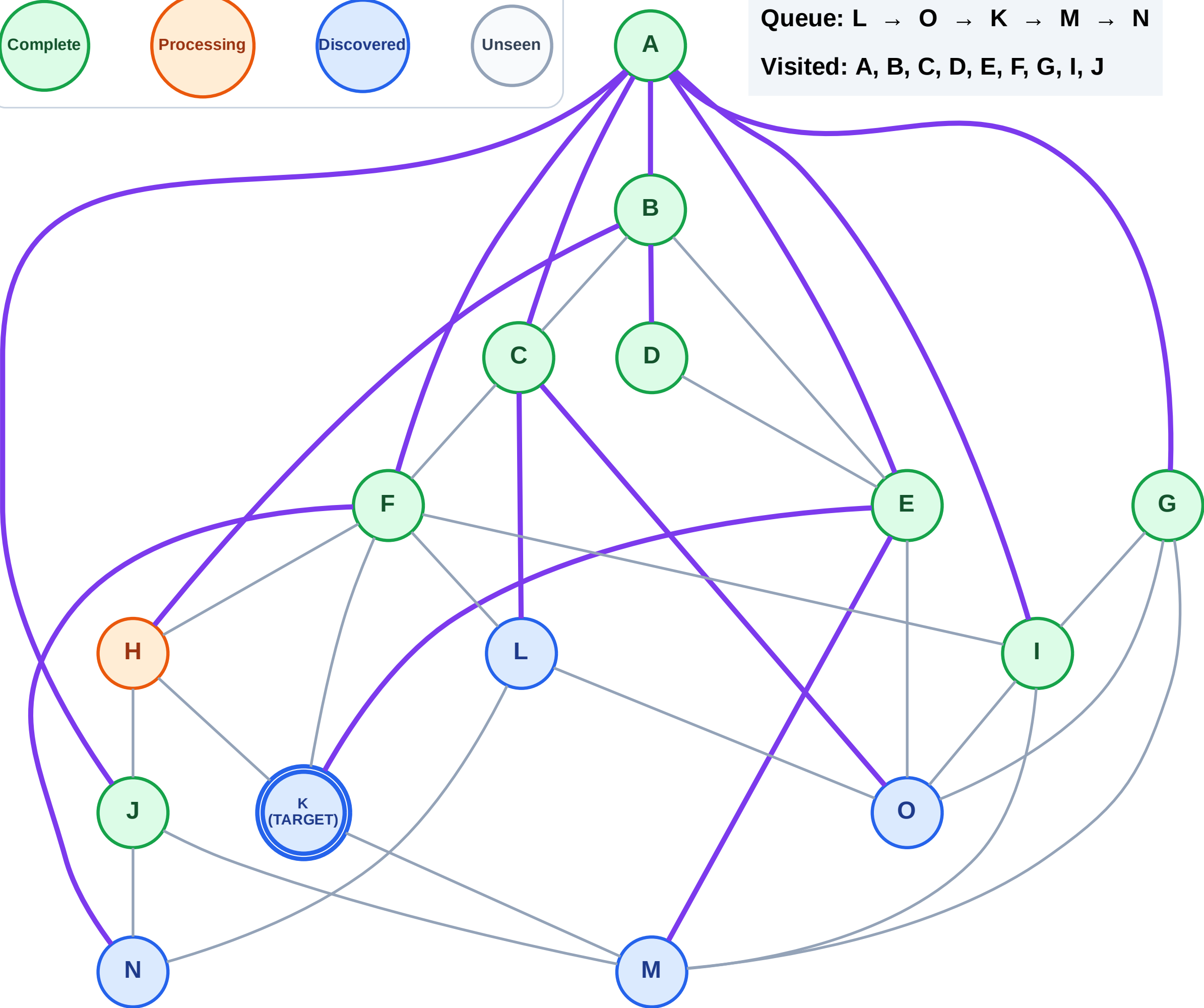
Processing

Discovered

Unseen

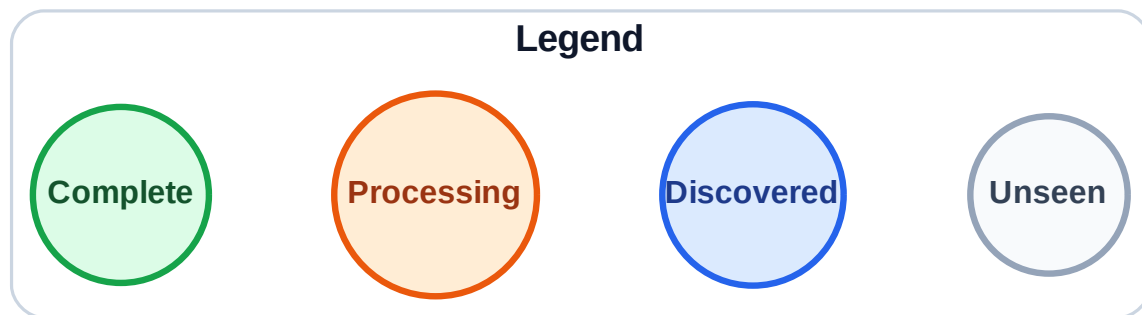
Queue: L → O → K → M → N

Visited: A, B, C, D, E, F, G, I, J



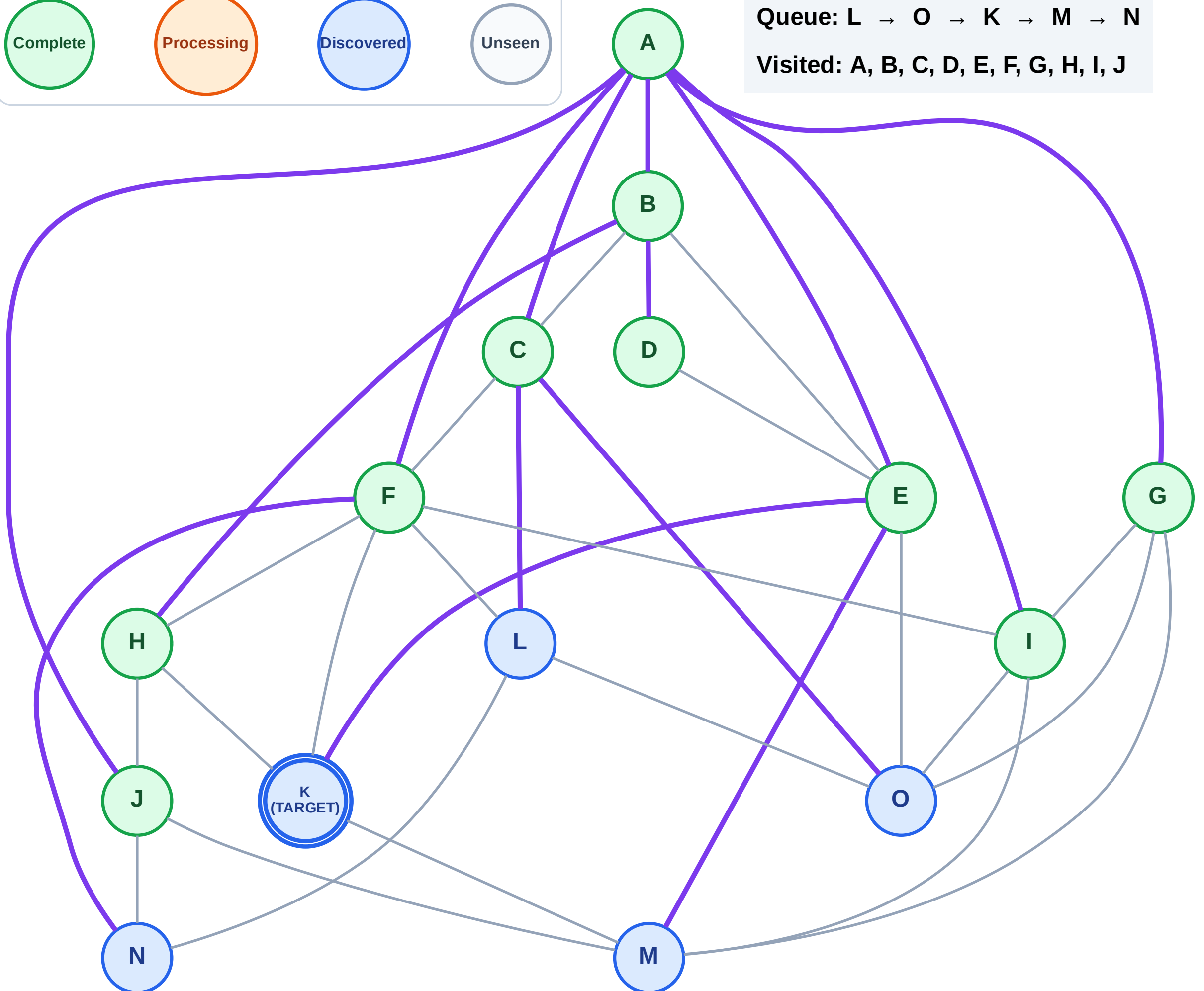
BFS Traversal

Step 21: No new neighbors from H



Queue: L → O → K → M → N

Visited: A, B, C, D, E, F, G, H, I, J



BFS Traversal

Step 22: Process node L

Legend

Complete

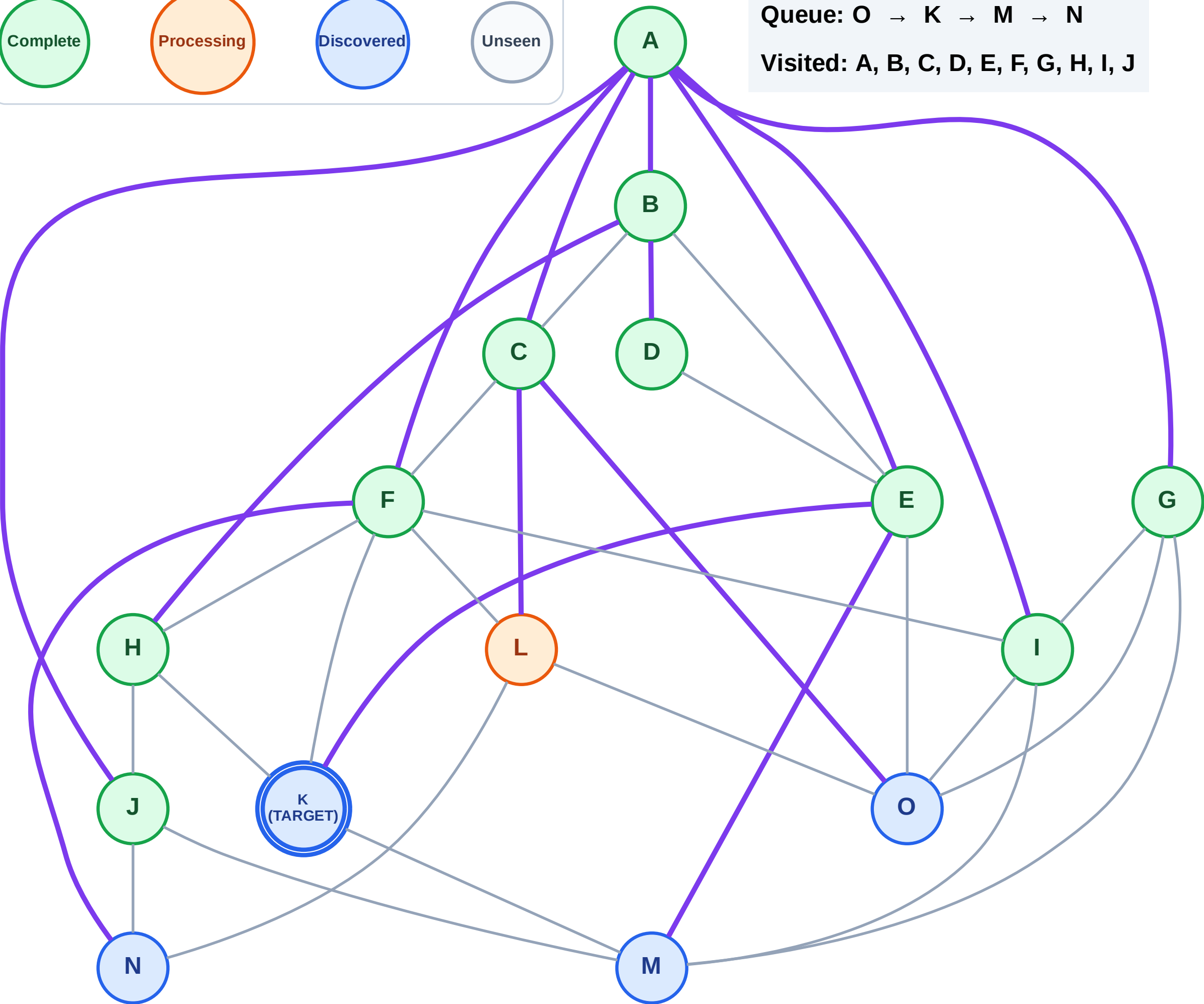
Processing

Discovered

Unseen

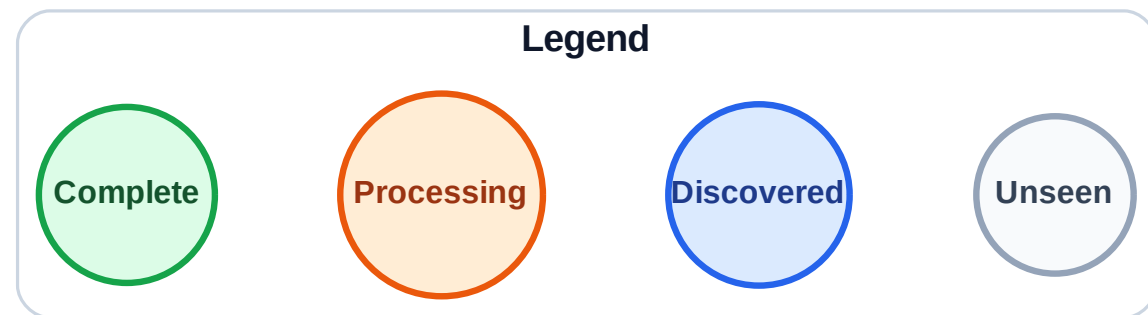
Queue: O → K → M → N

Visited: A, B, C, D, E, F, G, H, I, J



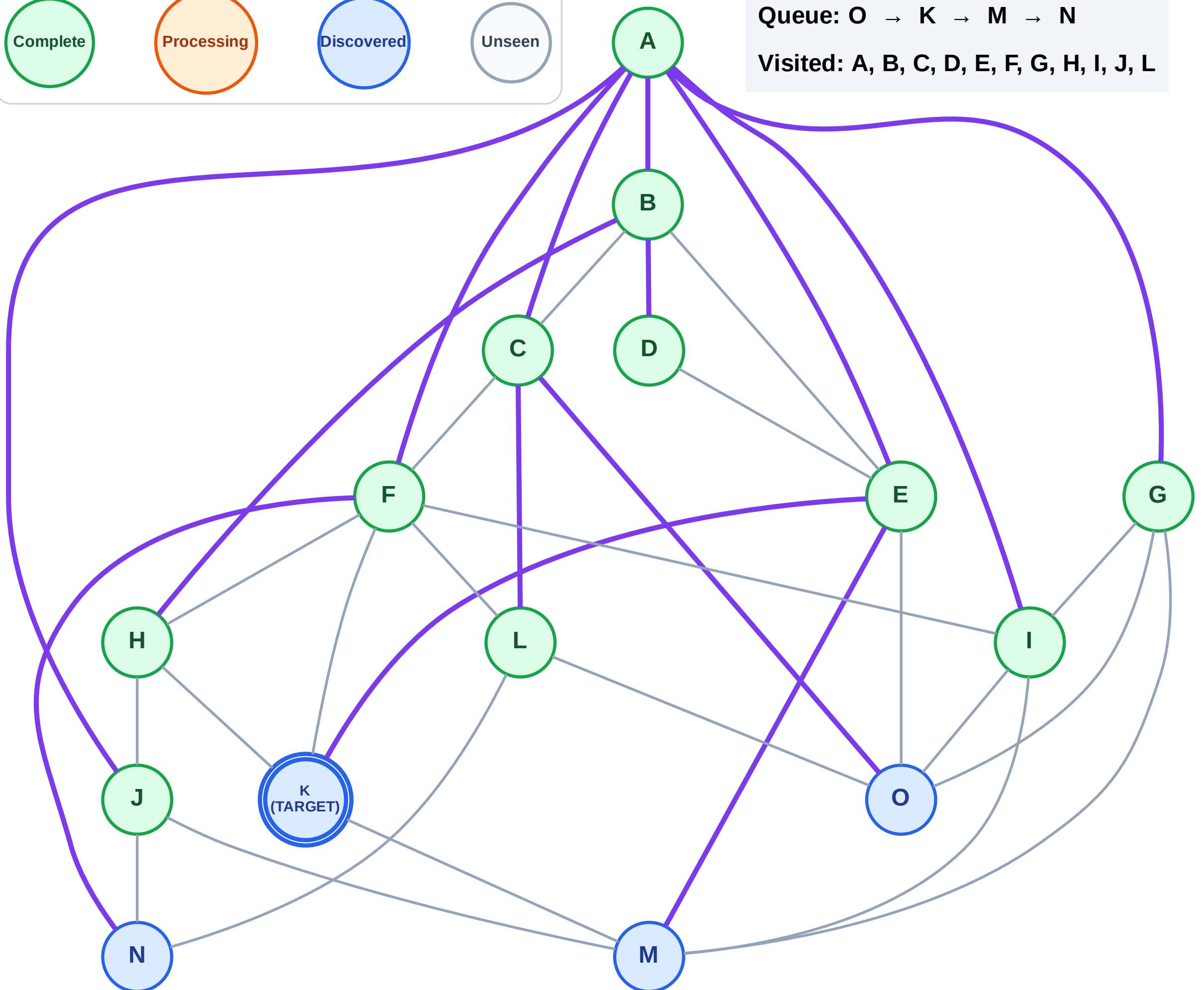
BFS Traversal

Step 23: No new neighbors from L



Queue: O → K → M → N

Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

Step 24: Process node O

Legend

Complete

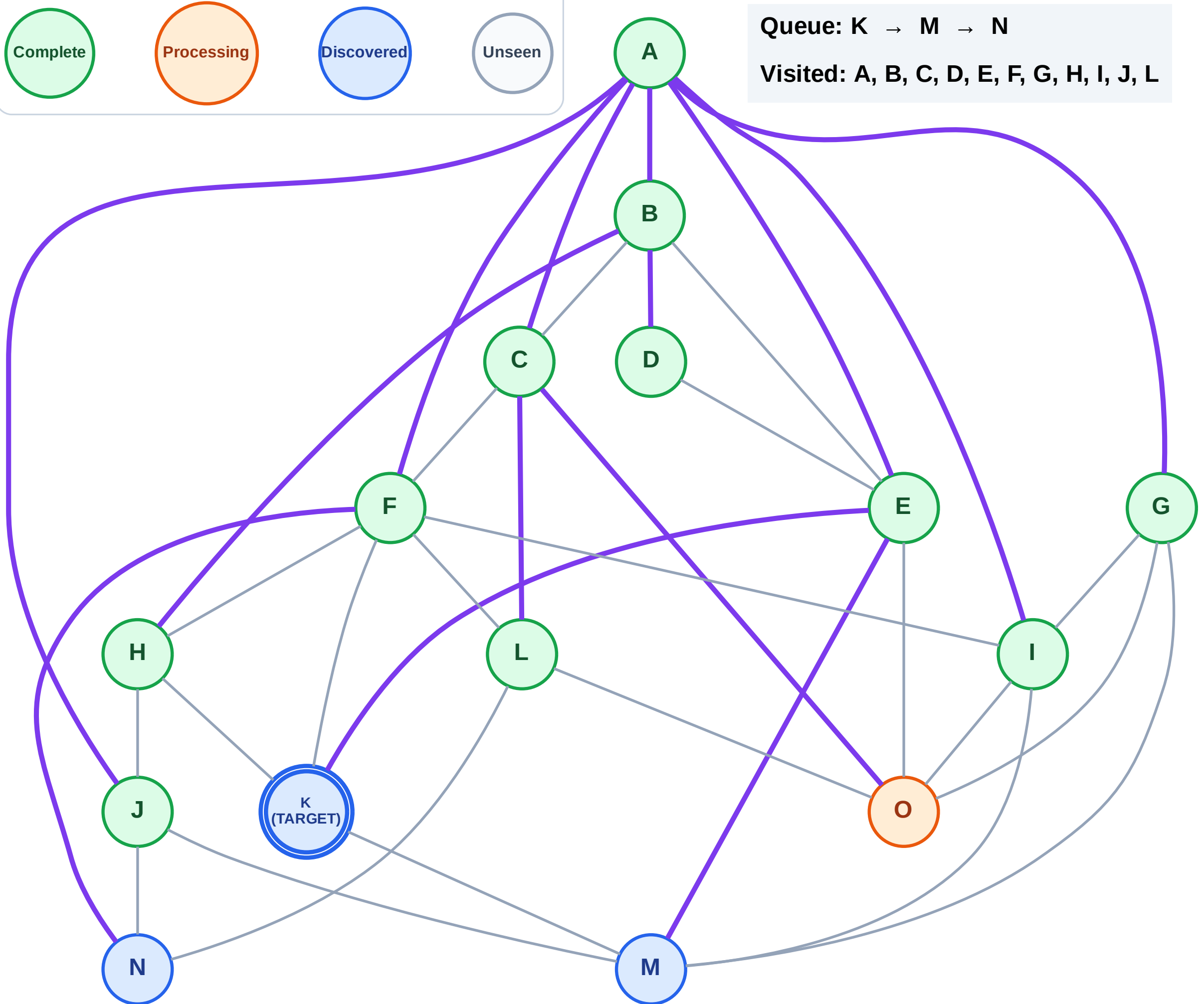
Processing

Discovered

Unseen

Queue: K → M → N

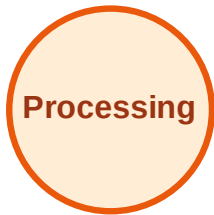
Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

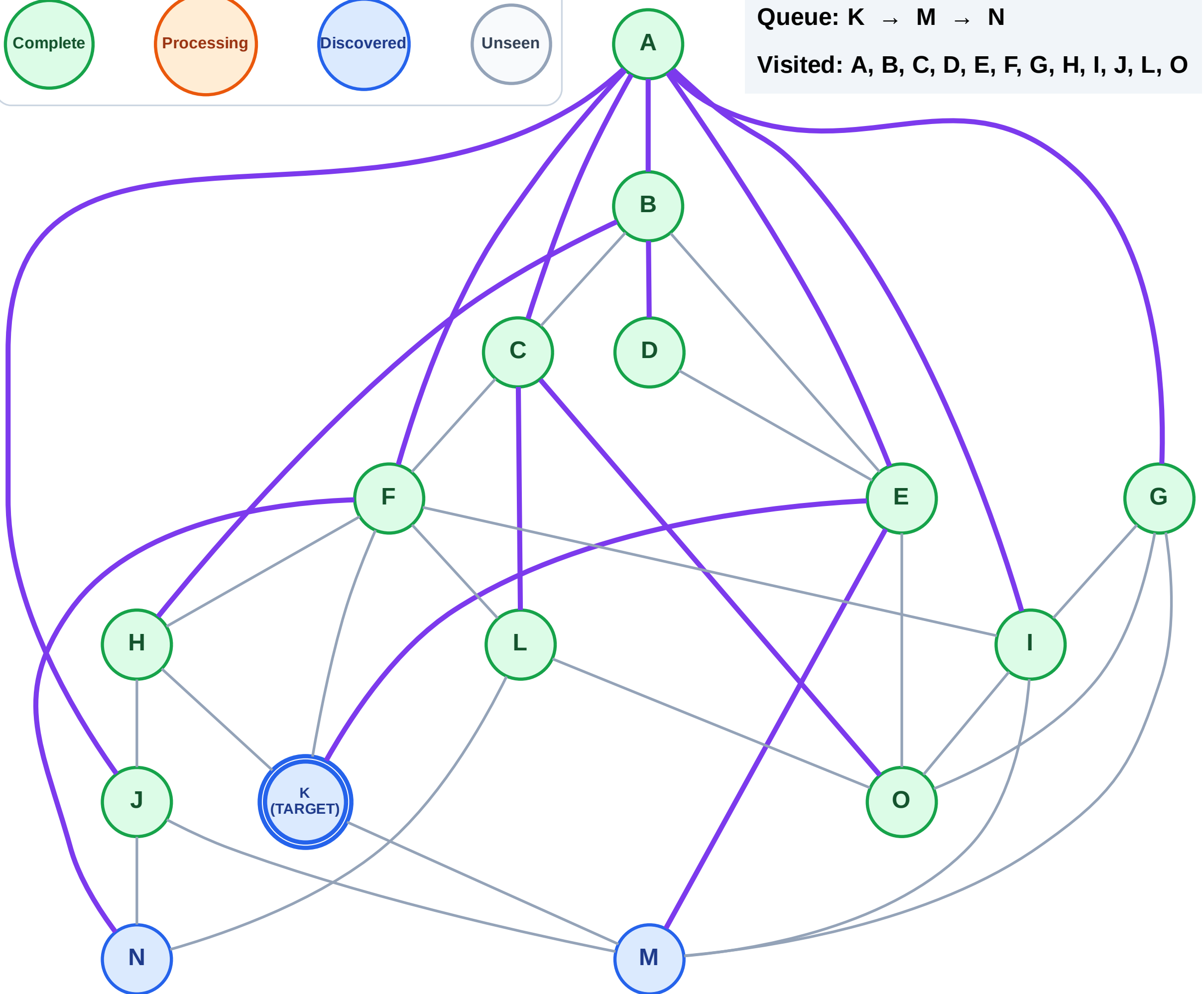
Step 25: No new neighbors from O

Legend



Queue: K → M → N

Visited: A, B, C, D, E, F, G, H, I, J, L, O



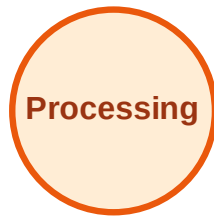
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



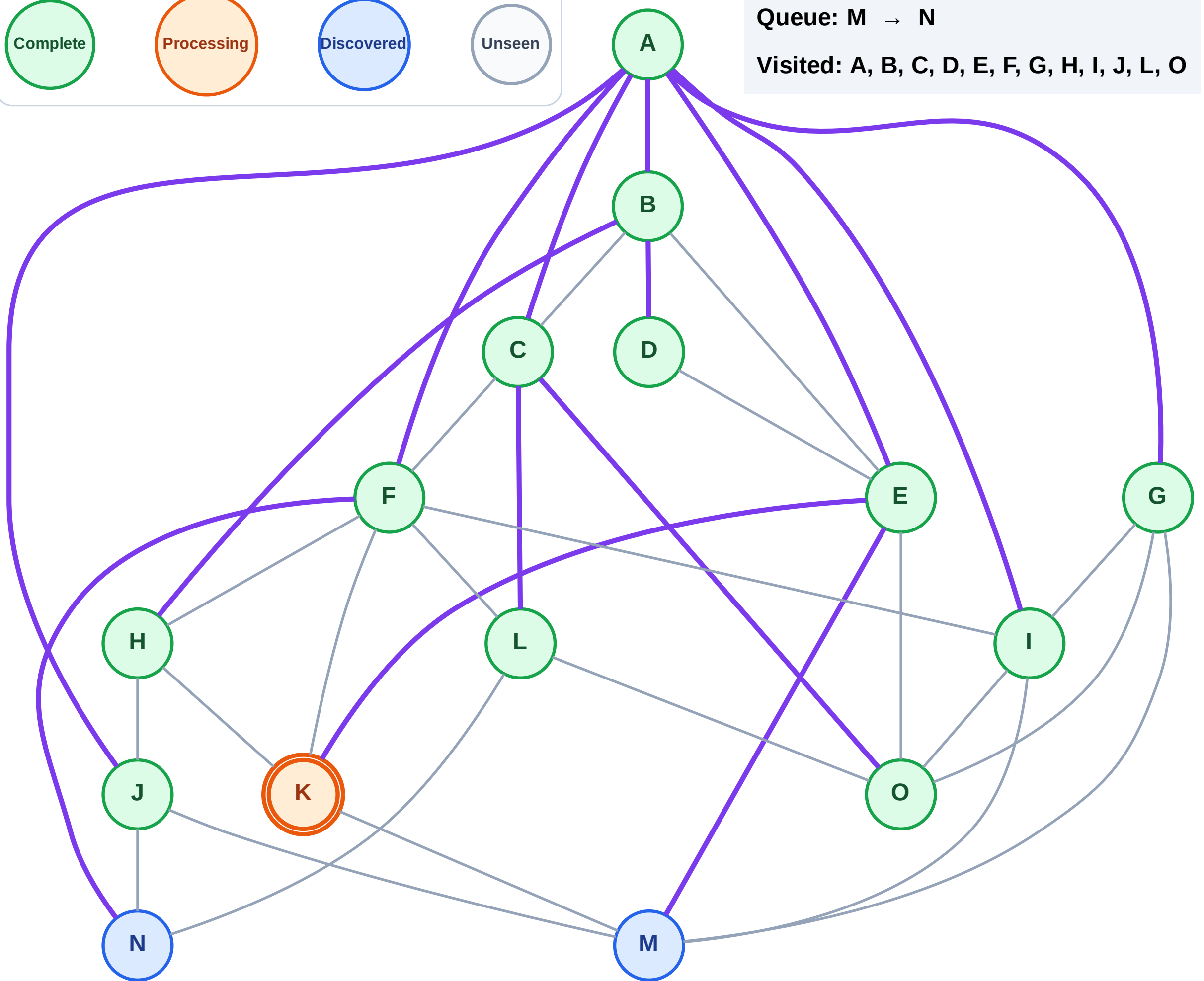
Discovered



Unseen

Queue: M → N

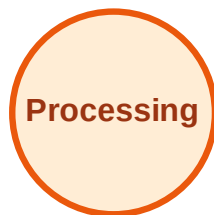
Visited: A, B, C, D, E, F, G, H, I, J, L, O



BFS Traversal

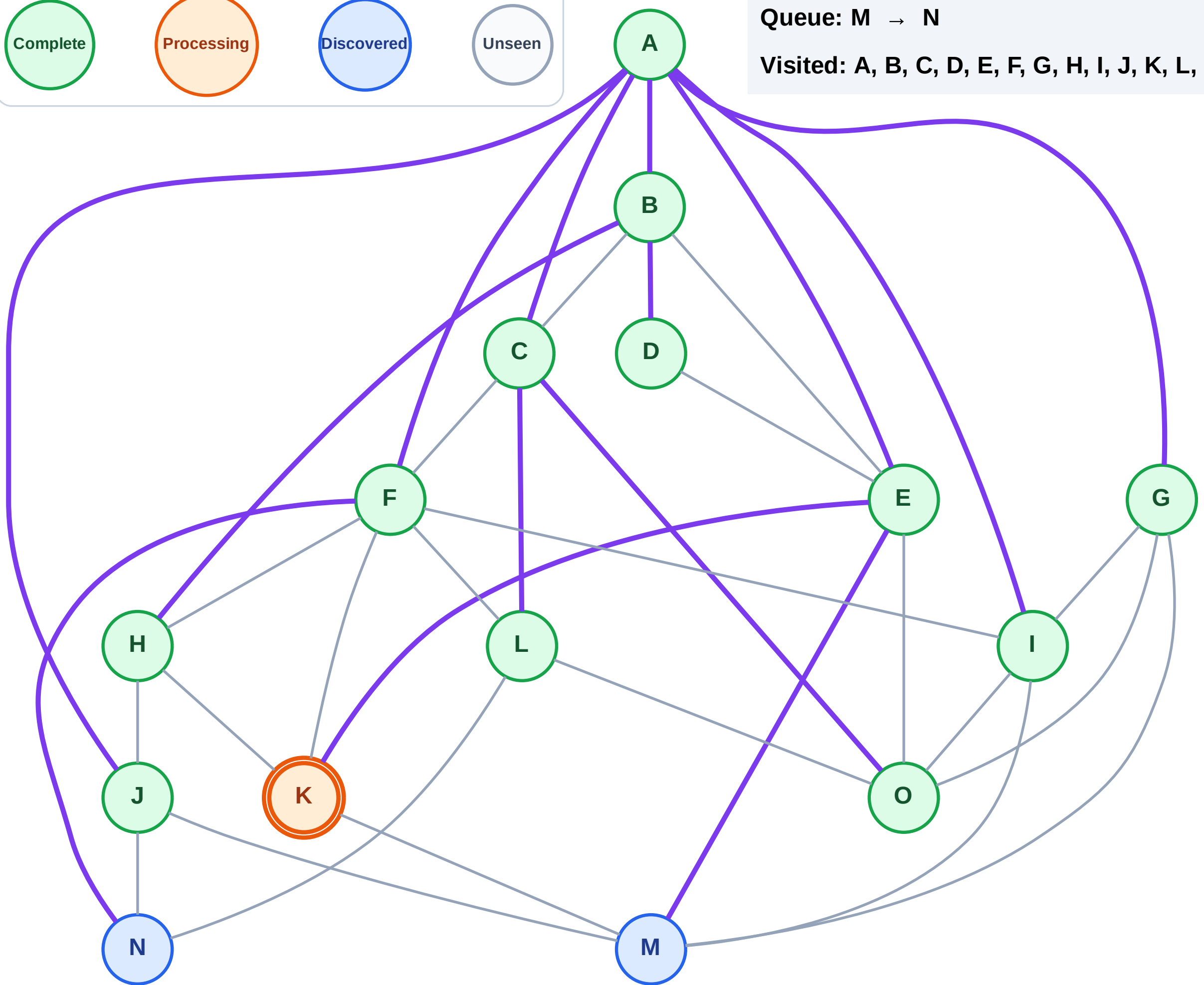
Step 27: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend



Queue: M → N

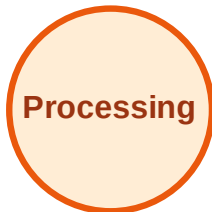
Visited: A, B, C, D, E, F, G, H, I, J, K, L, O



DFS Traversal

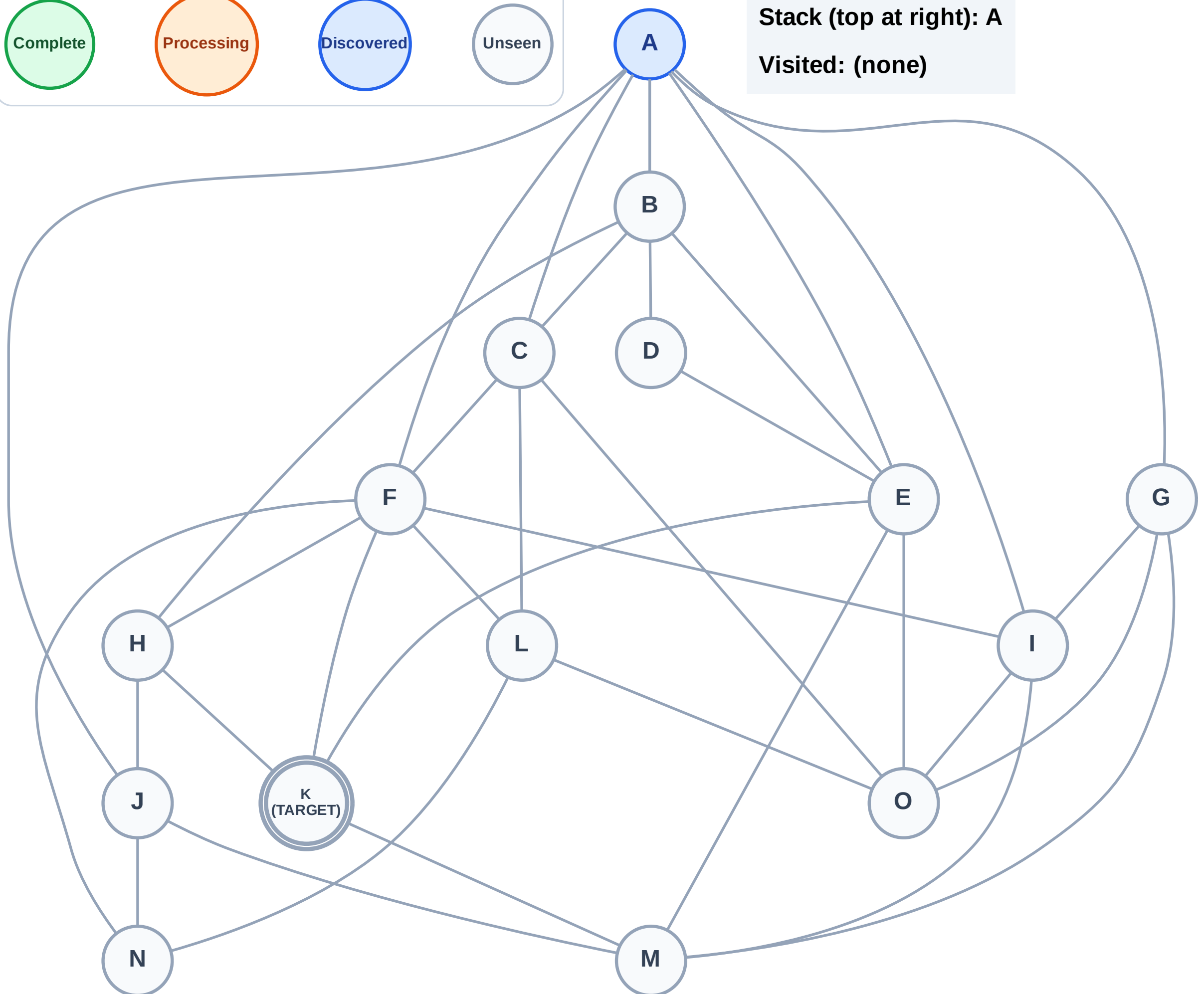
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

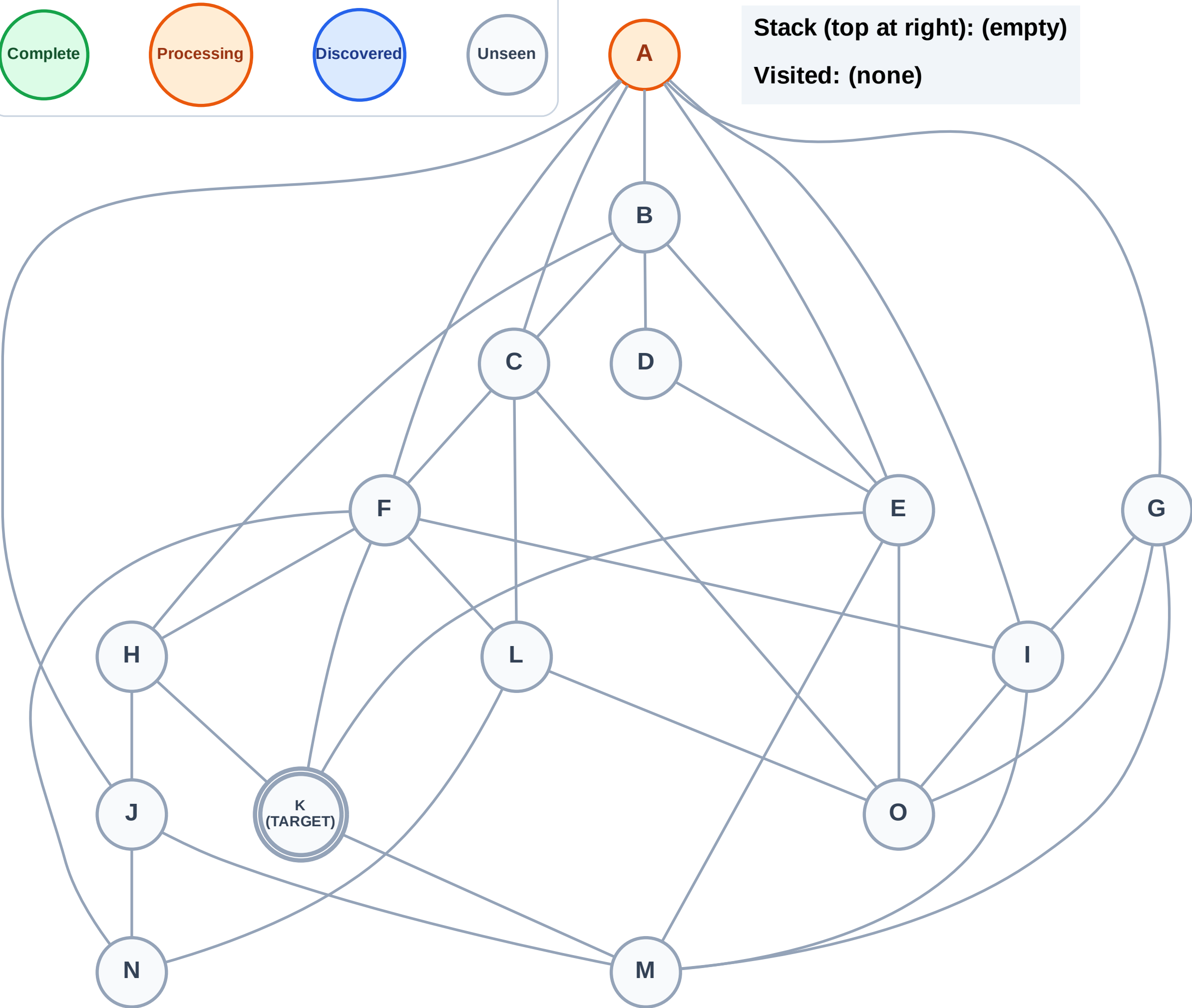
Processing

Discovered

Unseen

Stack (top at right): (empty)

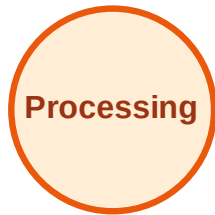
Visited: (none)



DFS Traversal

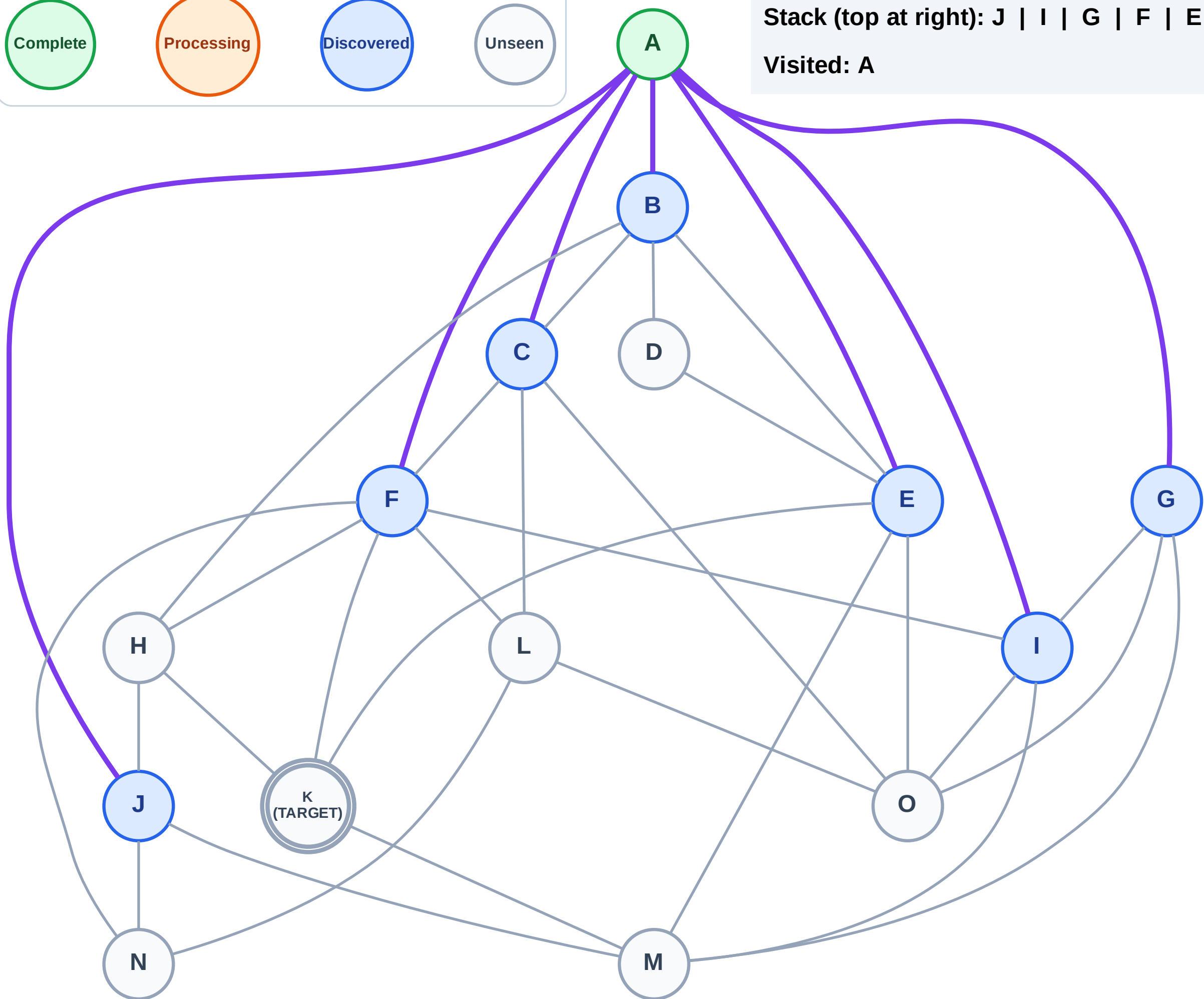
Step 3: Discover J, I, G, F, E, C, B from A

Legend



Stack (top at right): J | I | G | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

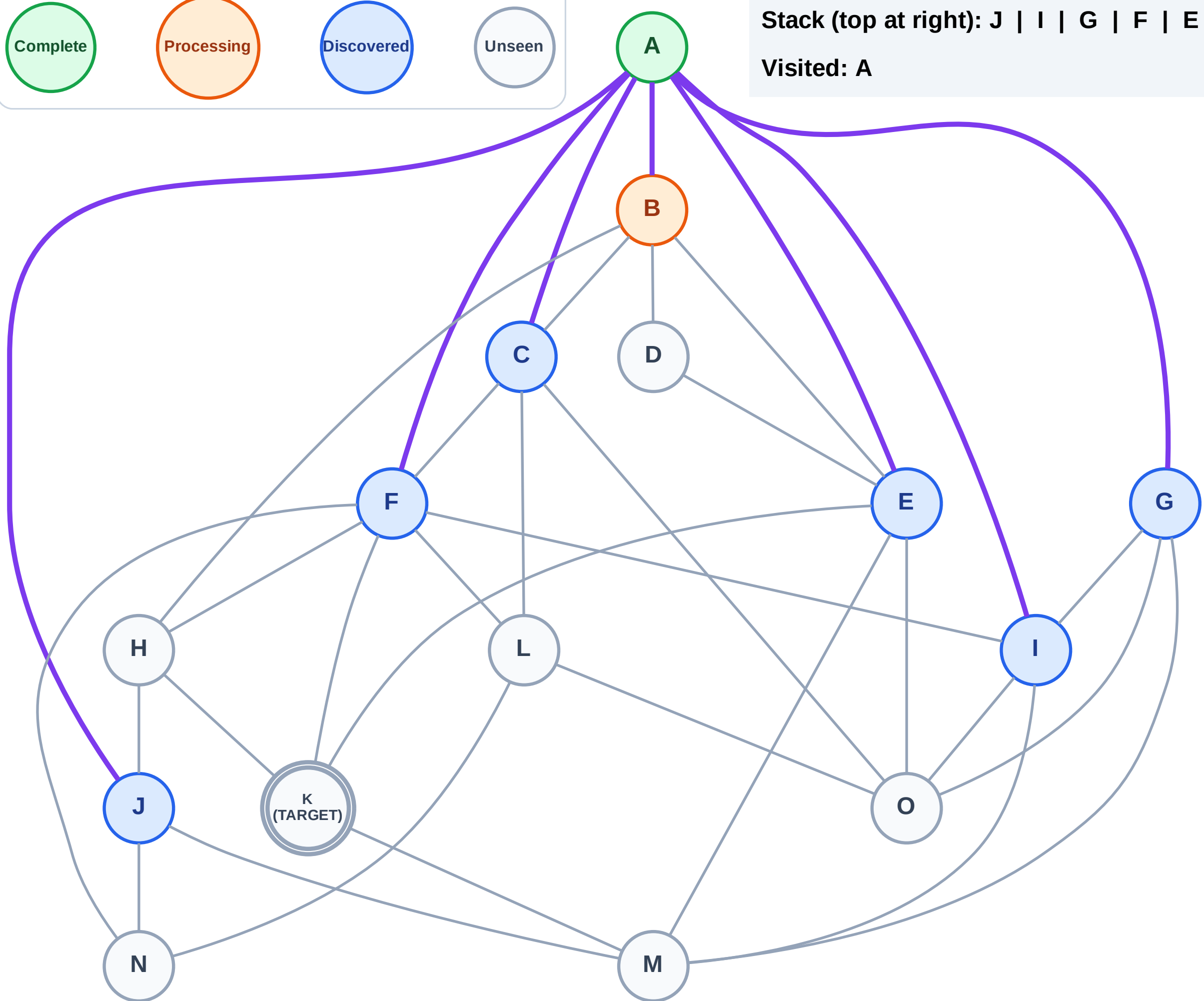
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | E | C

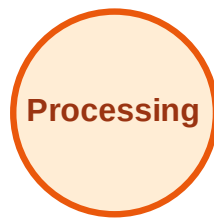
Visited: A



DFS Traversal

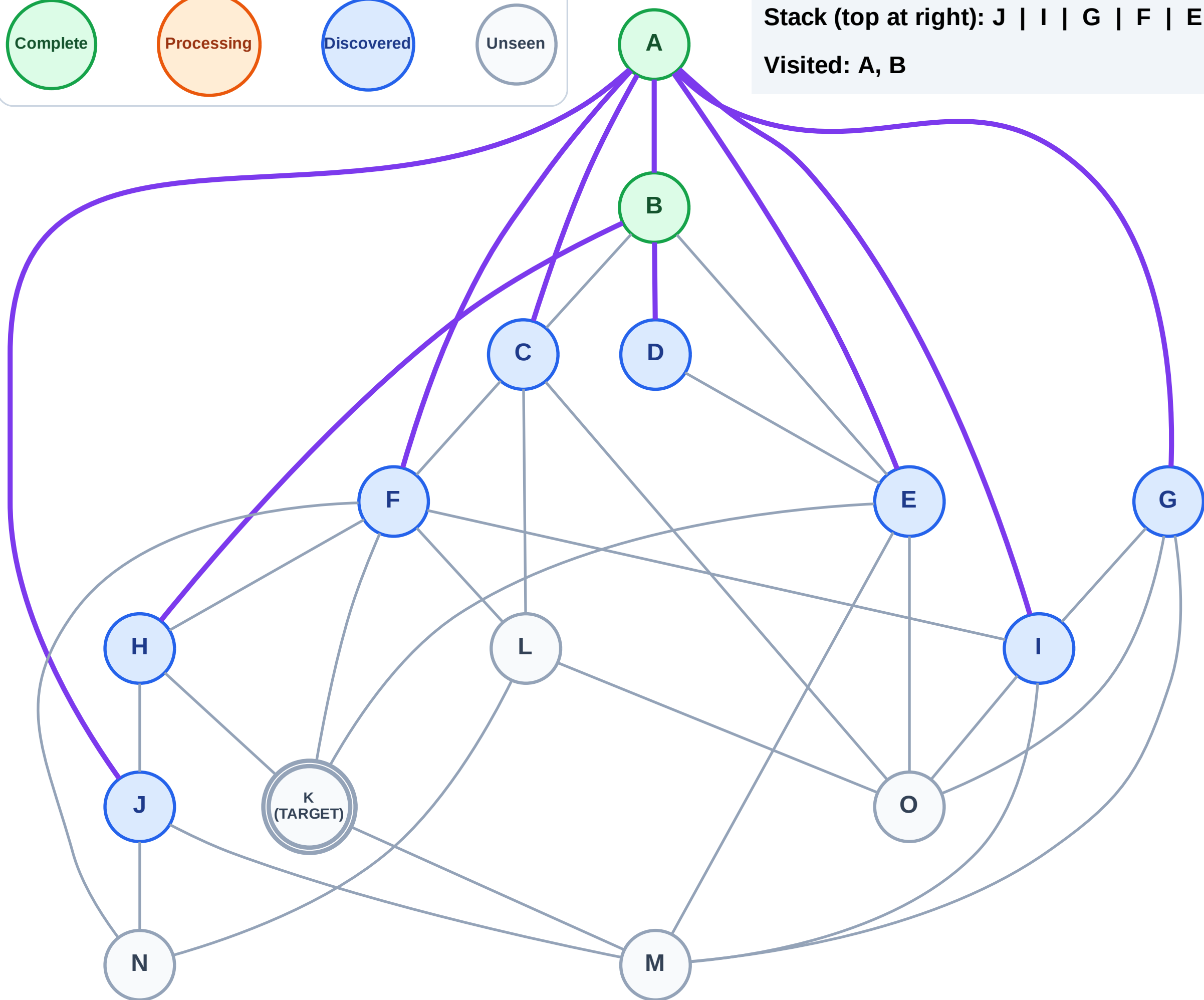
Step 5: Discover H, D from B

Legend



Stack (top at right): J | I | G | F | E | C | H | D

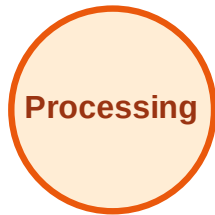
Visited: A, B



DFS Traversal

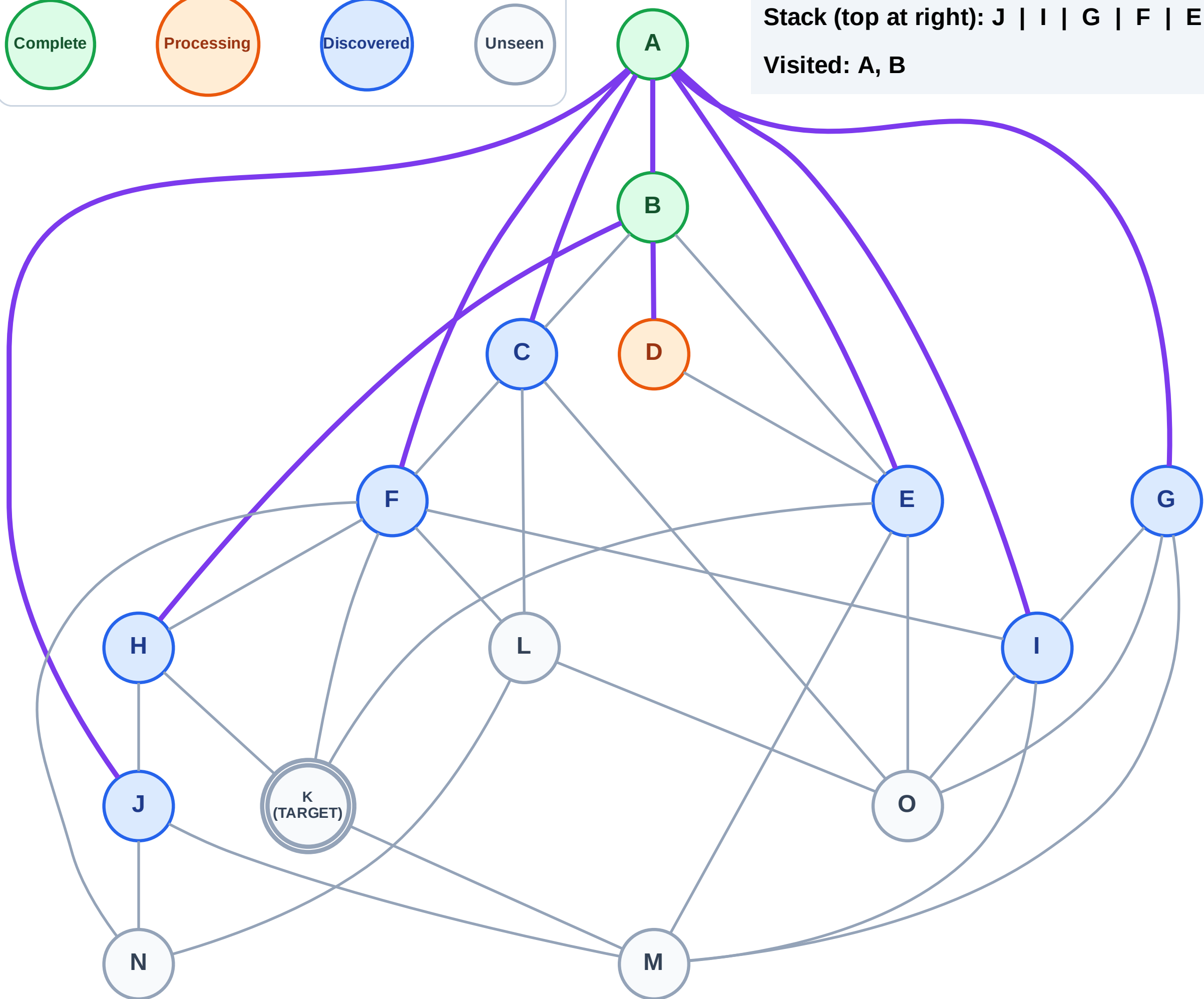
Step 6: Process node D

Legend



Stack (top at right): J | I | G | F | E | C | H

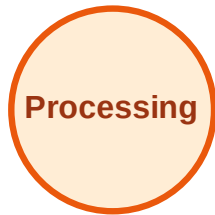
Visited: A, B



DFS Traversal

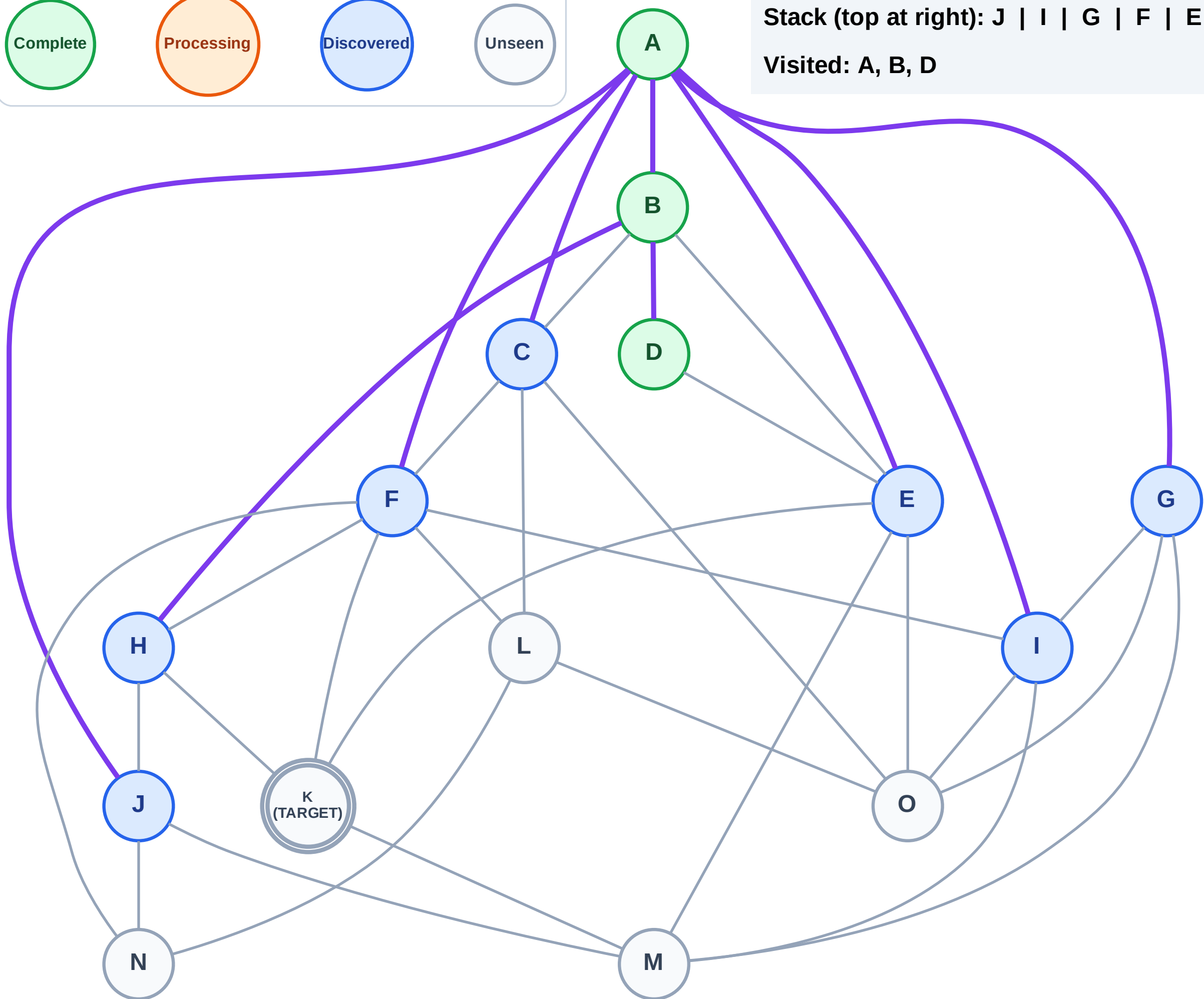
Step 7: No new neighbors from D

Legend



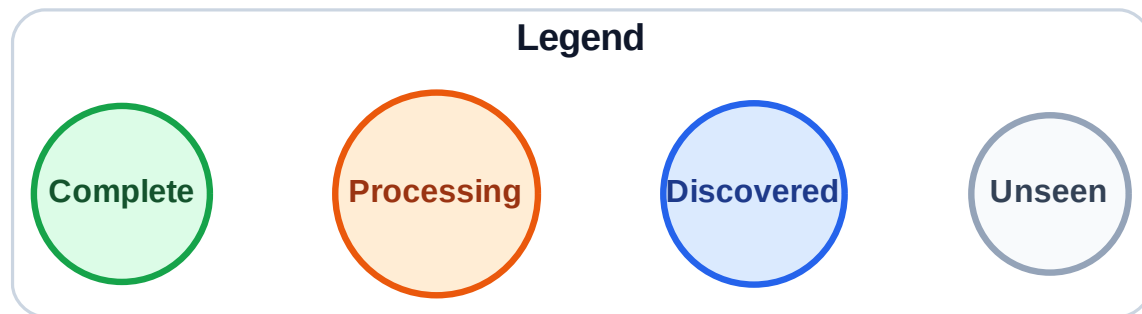
Stack (top at right): J | I | G | F | E | C | H

Visited: A, B, D

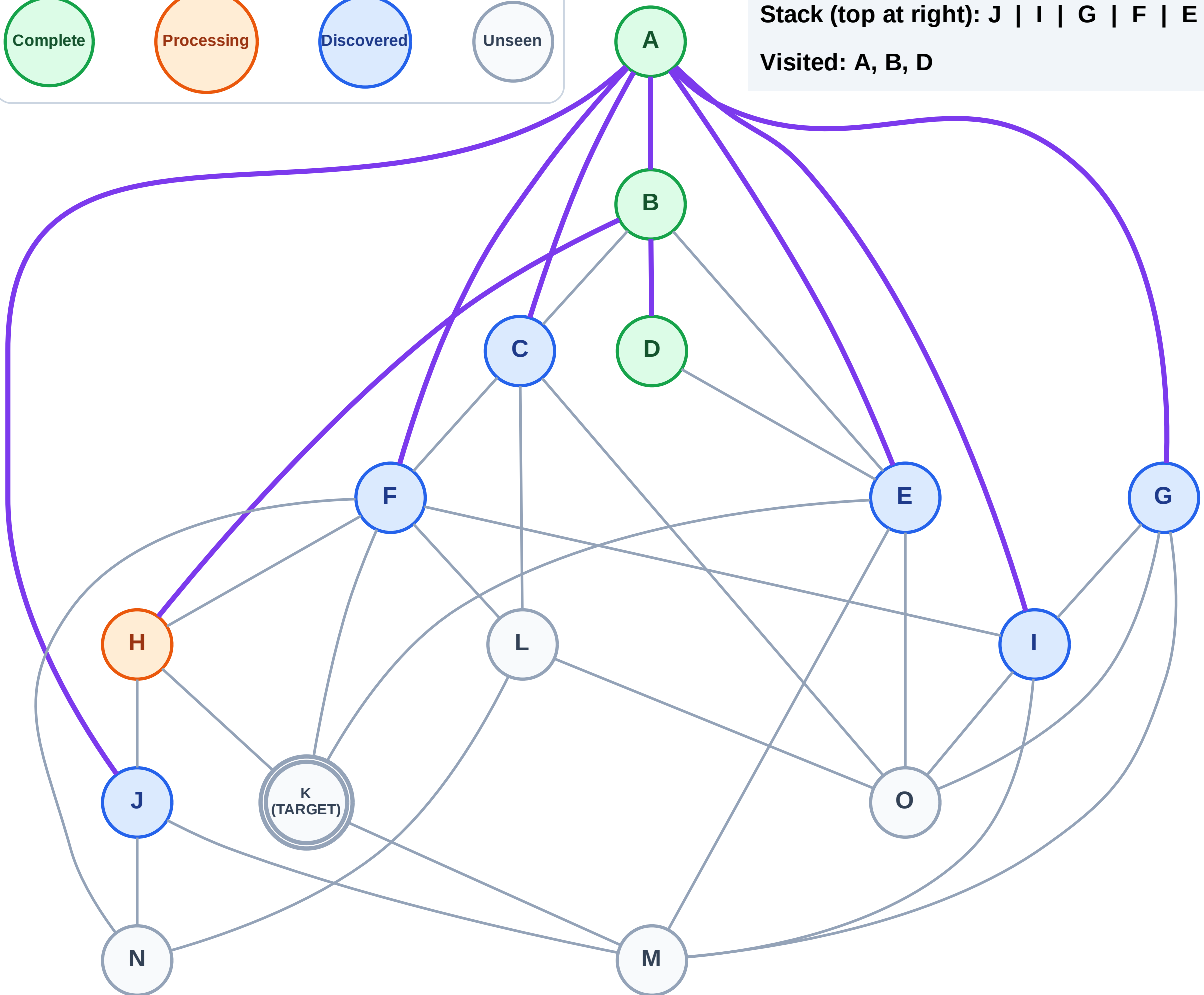


DFS Traversal

Step 8: Process node H



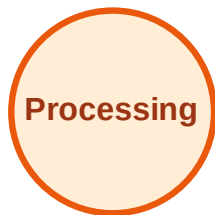
Stack (top at right): J | I | G | F | E | C
Visited: A, B, D



DFS Traversal

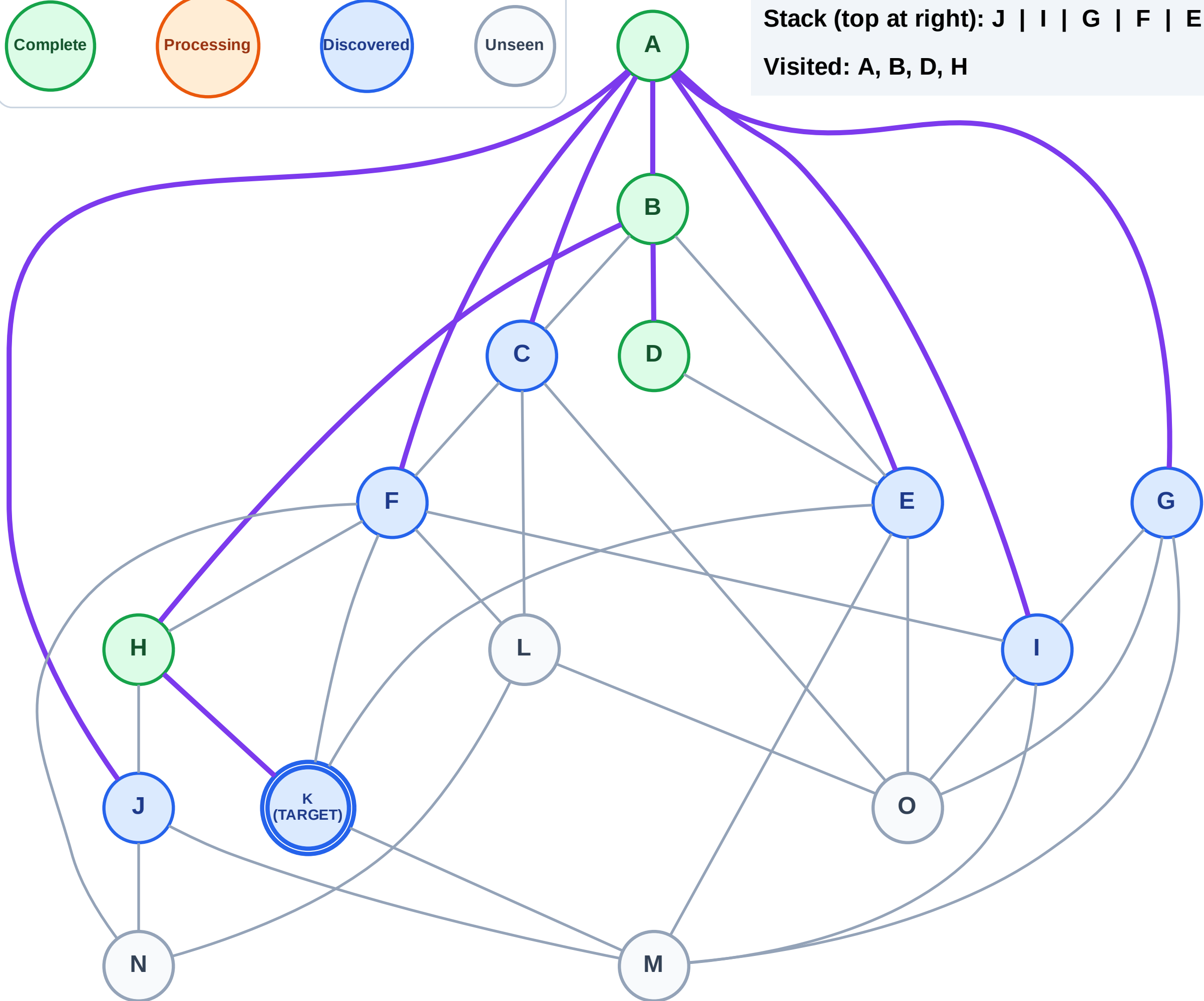
Step 9: Discover K from H

Legend



Stack (top at right): J | I | G | F | E | C | K

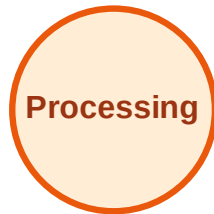
Visited: A, B, D, H



DFS Traversal

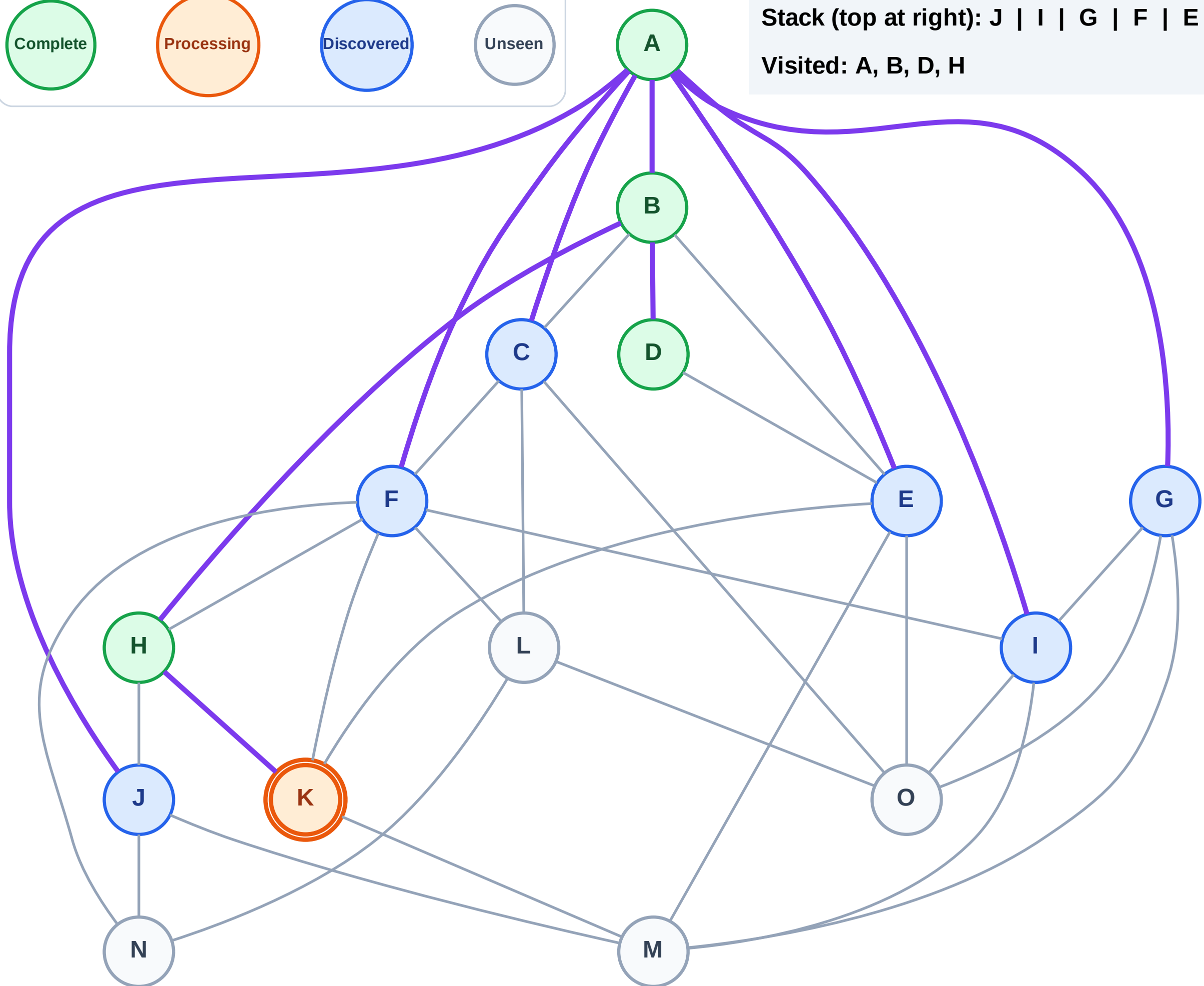
Step 10: Process node K

Legend



Stack (top at right): J | I | G | F | E | C

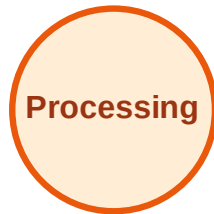
Visited: A, B, D, H



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend



Stack (top at right): J | I | G | F | E | C

Visited: A, B, D, H, K

